MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Two blocks of masses $10\ \text{kg}$ and $30\ \text{kg}$ are placed along a vertical line. If $10\ \text{kg}$ block is raised through a height of $7\ \text{cm}$, then the distance through which other mass should be moved to raise the center of mass of the system by $1\ \text{cm}$ is

(A) $1\ \text{cm}$ up

(B) $1\ \text{cm}$ down

(C) $2\ \text{cm}$ down

(D) $2\ \text{cm}$ up

Q.2

Statement 1: The plot of atomic number (y−axis) versus the number of neutrons (x−axis) for stable nuclei shows a curvature towards (x−axis) from the line of 45° slope as the atomic number is increased.
Statement 2: Proton-proton electrostatic repulsion begin to overcome attractive forces involving protons and neutrons in heavier nuclides.

(A) Statement 1 is true, statement 2 is true, statement 2 is correct explanation for the statement 1.

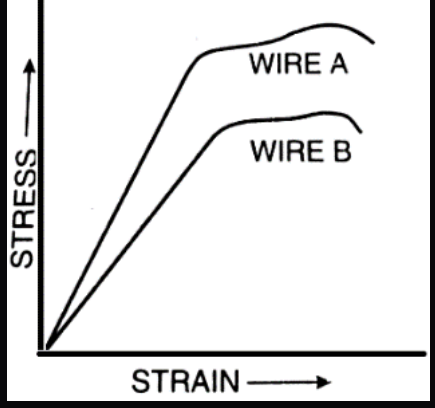
(B) Statement 1 is true, statement 2 is true, statement 2 is not a correct explanation for statement 1.

(C) Statement 1 is true and statement 2 is false.

(D) Statement 1 is false and statement 2 is true.

Q.3

The stress versus strain graphs for wires of two materials A and B are as shown in figure. Which material is more ductile?



(A) Material A

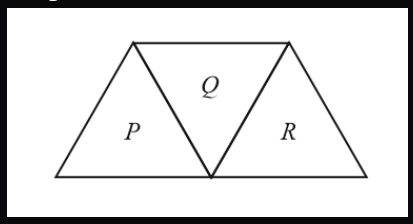
(B) Material B

(C) Both material A and material B

(D) None of these

Q.4

A given ray of light suffers minimum deviation in an equilateral prism P. Additional prisms Q and R of identical shape and of same material as P are now added as shown in the figure. The ray will now suffer



(A) greater deviation.

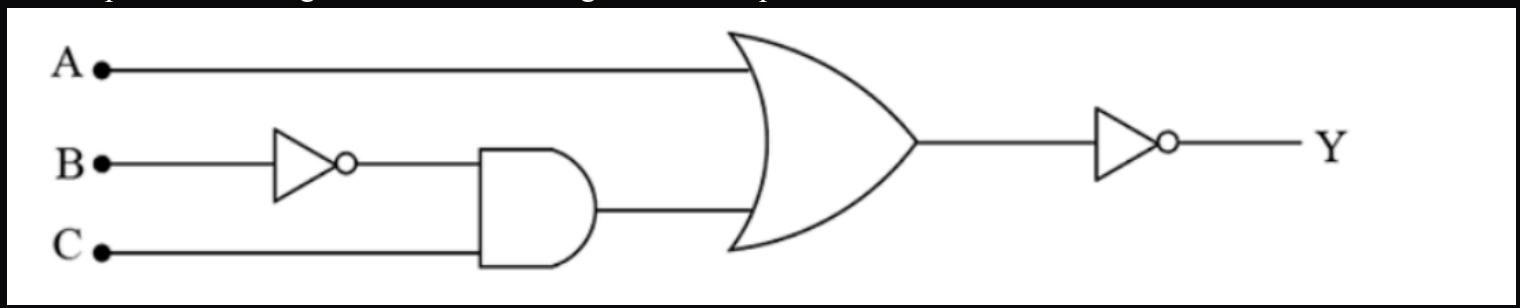
(B) no deviation.

(C) same deviation.

(D) total internal reflection.

Q.5

The output Y of the logic circuit shown in figure is best represented as:



(A) $\bar{A} + \overline{B \cdot C}$

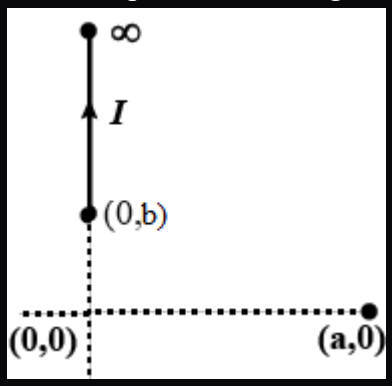
(B) $A + B \cdot C$

(C) $\overline{A + \overline{B} \cdot C}$

(D) $\overline{A + B \cdot C}$

Q.6

An infinitely long wire carrying current i is along Y-axis such that its one end is at point $(0, b)$ while the wire extends upto ∞. The magnitude of magnetic field strength at point $P(a, 0)$ is



(A) $\frac{\mu_0 i}{4\pi a}\left(1+\frac{b}{\sqrt{a^2+b^2}}\right)$

(B) $\frac{\mu_0 i}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$

(C) $\frac{\mu_0 i}{4\pi a}\left(1-\frac{a}{\sqrt{a^2+b^2}}\right)$

(D) $\frac{\mu_0 i}{4\pi a}\left(1+\frac{a}{\sqrt{a^2+b^2}}\right)$

Q.7

The correct match between the entries in column I and column II are :

	I		**II**
	Radiation		**Wavelength**
a	Microwave	i	$100\ m$
b	Gamma rays	ii	10^{-15}m
c	A.M. radio	iii	10^{-10}m
d	X–rays	iv	10^{-3}m

(A) (a)–(ii), (b) – (i), (c) – (iv), (d) – (iii)

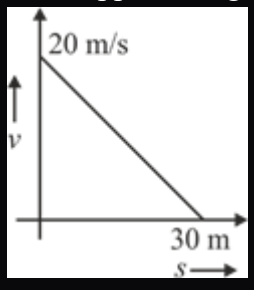
(B) (a)–(i), (b) – (iii), (c) – (iv), (d) – (ii)

(C) (a)–(iii), (b) – (ii), (c) – (i), (d) – (iv)

(D) (a)–(iv), (b) – (ii), (c) – (i), (d) – (iii)

Q.8

If the velocity v of a particle moving along a straight line decreases linearly with its displacement s is $20\ \text{ms}^{-1}$ to a value approaching to zero at $s = 30\ \text{m}$, then acceleration of the particle at $s = 15\ \text{m}$, is



(A) $\frac{2}{3}\ \mathrm{ms}^{-2}$

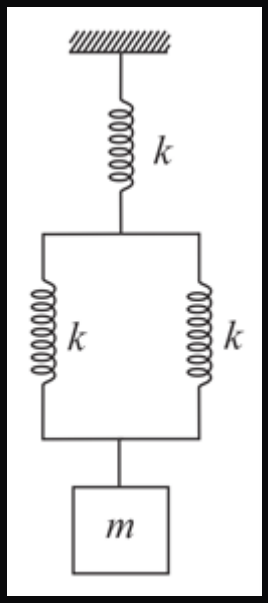
(B) $-\frac{2}{3}\ \mathrm{ms}^{-2}$

(C) $\frac{20}{3}\ \mathrm{ms}^{-2}$

(D) $-\frac{20}{3}\ \mathrm{ms}^{-2}$

Q.9

Find the time period of the body (mass m), shown in the figure, when displaced slightly in the vertical direction.



(A) $2\pi\sqrt{\frac{m}{2k}}$

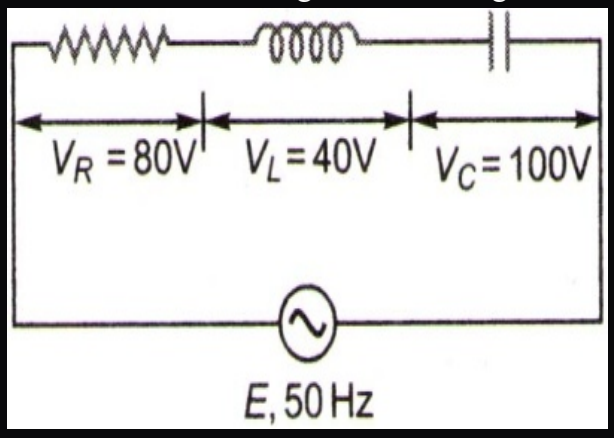
(B) $2\pi\sqrt{\frac{3m}{k}}$

(C) $2\pi\sqrt{\frac{3m}{2k}}$

(D) $\pi\sqrt{\frac{3m}{k}}$

Q.10

The value of alternating emf E in the given circuit will be



(A) 220 V

(B) 140 V

(C) 100 V

(D) 20 V

Q.11

According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal vs the frequency of the incident radiation gives a straight line whose slope

(A) depends on the nature of the metal used

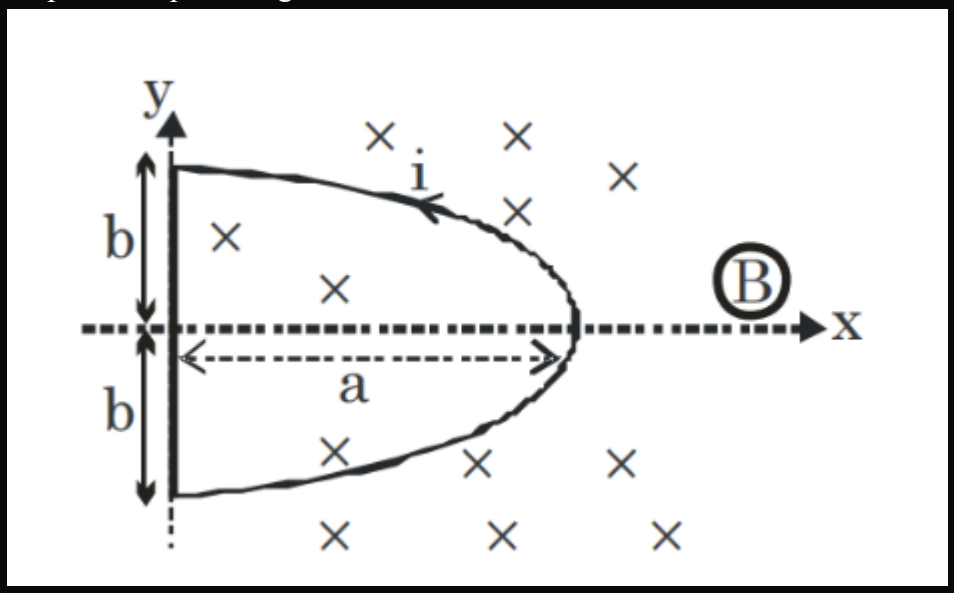
(B) depends on the intensity of the radiation

(C) depends both on the intensity of the radiation and the metal use

(D) is the same for all metals and independent of the intensity of the radiation

Q.12

In the figure, there is a conducting wire having current i and which has a shape of closed half ellipse $\left[\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\right]$ is kept in a uniform magnetic field B as shown. The magnitude of magnetic dipole moment of loop and torque acting on it are-



(A) π abi, 0

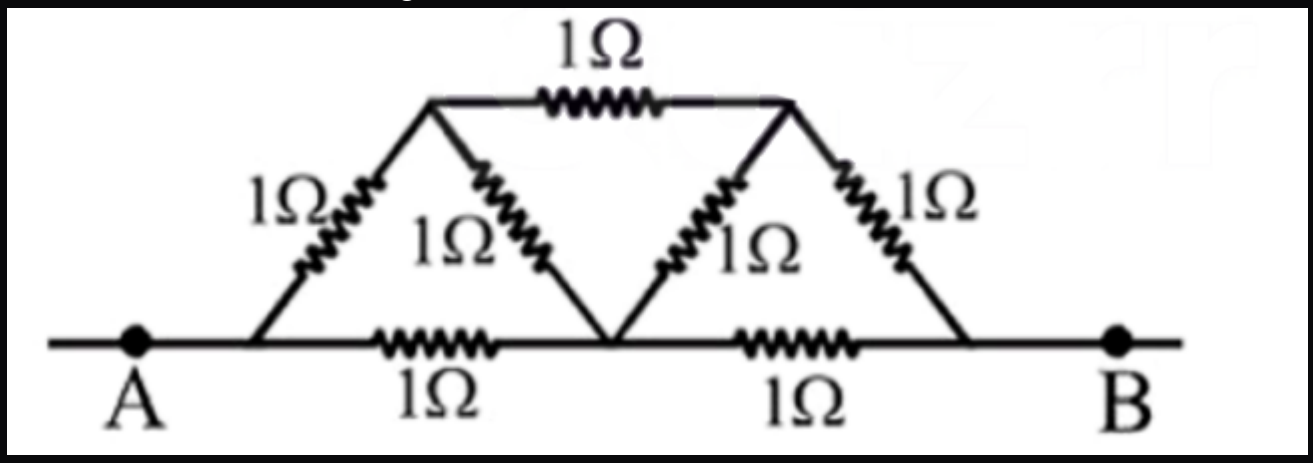
(B) πab i, $\pi abiB$

(C) $\frac{\pi abi}{2}, \frac{\pi abiB}{2}$

(D) $\frac{\pi abi}{2}, 0$

Q.13

In the network shown in the Fig, each resistance is 1Ω. The effective resistance between A and B is



(A) $\frac{4}{3}\Omega$

(B) $\frac{3}{2}\Omega$

(C) 7Ω

(D) $\frac{8}{7}\Omega$

Q.14

Assertion: Consider a system of gas having N molecules; Instantaneous K.E. of few molecules can be greater than average K.E. of the molecules of the given gas.
Reason: Number of molecules having most probable speed is greater than number of molecules having average speed.

(A) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(B) Both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

(C) Assertion is true but the Reason is false.

(D) Assertion is false but Reason is true.

Q.15

Match the physical quantities with their correct SI units.

List-I		List-II	
(A)	Planck's constant	(I)	$\mathrm{kg\ m\ s^{-2}}$
(B)	Torque	(II)	$\mathrm{kg^{-1}\ m^3}$
(C)	Weight	(III)	$\mathrm{m^2\ kg\ s^{-1}}$
(D)	Specific volume	(IV)	$\mathrm{kg\ m^2\ s^{-2}}$

The correct match is

(A) $\mathrm{A = II, B = IV, C = I, D = III}$

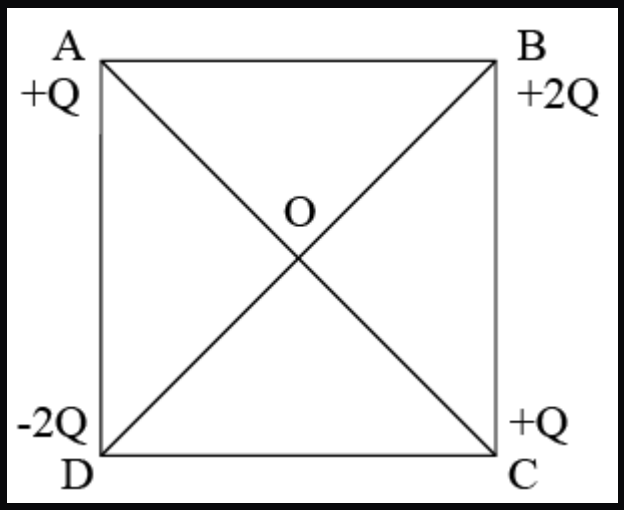
(B) $\mathrm{A = III, B = IV, C = I, D = II}$

(C) $\mathrm{A = IV, B = I, C = II, D = III}$

(D) $\mathrm{A = II, B = III, C = I, D = IV}$

Q.16

Four charges are arranged at the corners of a square $ABCD$ as shown in the figure. The force on the charge kept at the centre O is



(A) zero

(B) along the diagonal AC

(C) along the diagonal BD

(D) along perpendicular to the side AB

Q.17

If R is the radius of the Earth then the height above the Earth's surface at which the acceleration due to gravity decreases by 20% is

(A) $\left(\frac{\sqrt{5}}{2}-1\right)R$

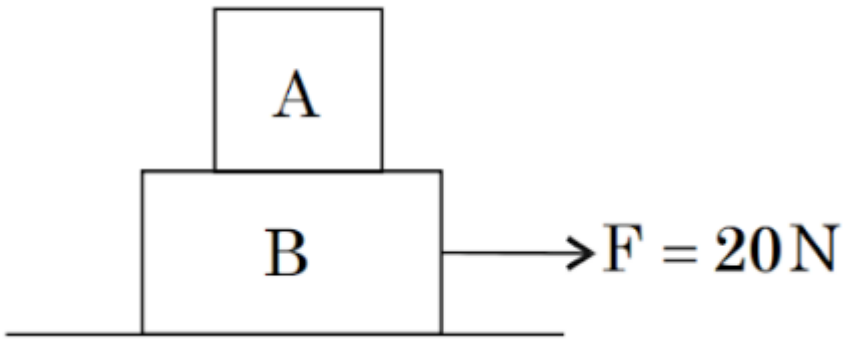
(B) $\left(\frac{\sqrt{5}}{2}+1\right)R$

(C) $(5\sqrt{2}-1)R$

(D) $(5\sqrt{2}+1)R$

Q.18

Block A of mass $2\ \text{kg}$ is placed over block B of mass $8\ \text{kg}$. The combination is placed over a smooth horizontal surface. Coefficient of friction between A and B is 0.5. A horizontal force of $20\ \text{N}$ is applied on block B. The force of friction on A is



(A) 10 N

(B) 8 N

(C) 6 N

(D) 4 N

Q.19

If E_P and E_K are the potential energy and kinetic energy of the electron in stationary orbit in the hydrogen atom, the value of $\frac{E_p}{E_K}$ is

(A) 2

(B) -1

(C) 1

(D) -2

Q.20

One mole of an ideal gas under goes a process $P = P_0\left[1 + \left(\frac{2V_0}{V}\right)^2\right]^{-1}$ where P_0, V_0 are constants. Change in temperature of the gas when volume is changed from $V = V_0$ to $V = 2V_0$ is:

(A) $\frac{4}{5}\frac{P_0V_0}{R}$

(B) $\frac{3}{4}\frac{P_0V_0}{R}$

(C) $\frac{2}{3}\frac{P_0V_0}{R}$

(D) $\frac{9}{7}\frac{P_0V_0}{R}$

Q.21

The densities of two solid spheres A and B of the same radii R vary with radial distance r as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where k is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B, respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of n is

(A) Submit

Q.22

A block of mass $2\ \text{kg}$ is kept at the origin at $t = 0$ and is having velocity $4\sqrt{5}\ \text{m s}^{-1}$ in the positive x direction. The only force on it is a conservative and its potential energy is defined as $U = -x^3 + 6x^2 + 15$.
Its velocity when the force acting on it is zero (after the time $t = 0$) is - (mark answer to nearest integer)

(A) Submit

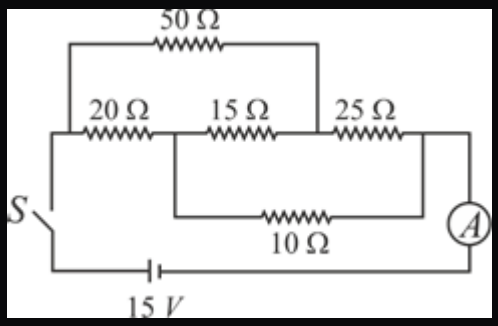
Q.23

A beam of light of wavelength $600\ \text{nm}$ from a monochromatic source falls on a single slit $1\ \text{mm}$ wide and the resulting diffraction pattern is observed on a screen $2\ \text{m}$ away. The distance between the first minima on either side of the central maxima is d (mm), find 10d

(A) Submit

Q.24

When switch S is closed, then the reading of ammeter is A, then value of $10A$ is _____



(A) Submit

Q.25

A particle of mass 1 mg and charge q is lying at the mid-point of two stationary particles kept at a distance 2 m when each is carrying same charge q. If the free charged particle is displaced from its equilibrium position through distance x ($x << 1$ m). The particle executes SHM. Its angular frequency of oscillation will be _____ $\times 10^8$ rad s^{-1} (if $q^2 = 10C^2$)

(A) Submit

Q.26

2-Butyne can be obtained by:
I. Dehydrogenation of 2 -butene.
II. Reaction of $1,1,1$-trichloroethane with Ag powder.
III. 1 equivalent of ethyne treated with excess of sodium followed by CH_3Cl.

(A) Both I and II

(B) Both II & III

(C) I, II & III

(D) Both I and III

Q.27

For the cell $\mathrm{Zn\ (s)|ZnSO_4\ (aq)||CuSO_4\ (aq)|\ Cu\ (s)}$, when the concentration of $\mathrm{Zn^{2+}}$ is 10 times the concentration of $\mathrm{Cu^{2+}}$, the expression for $\Delta \mathrm{G}\ \left(\mathrm{in\ J\ mol^{-1}}\right)$ is
[F =Faraday's constant, R =universal gas constant, T =temperature, $\mathrm{E^{o}cell} = 1.1\mathrm{V}$]

(A) $2.303 \times \mathrm{RT} + 1.1\ \mathrm{F}$

(B) $2.303 \times \mathrm{RT} - 2.2\ \mathrm{F}$

(C) $1.1\ \mathrm{F}$

(D) $-2.2\ \mathrm{F}$

Q.28

An organic compound contains 49.3% carbon, 6.84% hydrogen and its vapour density is 73 . Molecular formula of the compound is :

(A) $\mathrm{C_3H_5O_2}$

(B) $\mathrm{C_4H_{10}O_2}$

(C) $\mathrm{C_6H_{10}O_4}$

(D) $\mathrm{C_3H_{10}O_2}$

Q.29

Calculate the percentage of all the monochlorinated products obtained from the chlorination of 2 methyl butane. The relative reactivity of $1^{\circ}, 2^{\circ}$, and 3° hydrogen to chlorination is $(1 : 3.8 : 5)$.

(A) $15\%, 45\%, 20\%, 25\%$

(B) $14\%, 35\%, 23\%, 28\%$

(C) $12\%, 30\%, 22\%, 29\%$

(D) $17\%, 36\%, 20\%, 23\%$

Q.30

Assertion : A mixture of plant pigments can be separated by chromatography.
Reason: Chromatography is used for the separation of coloured substances into individual components.

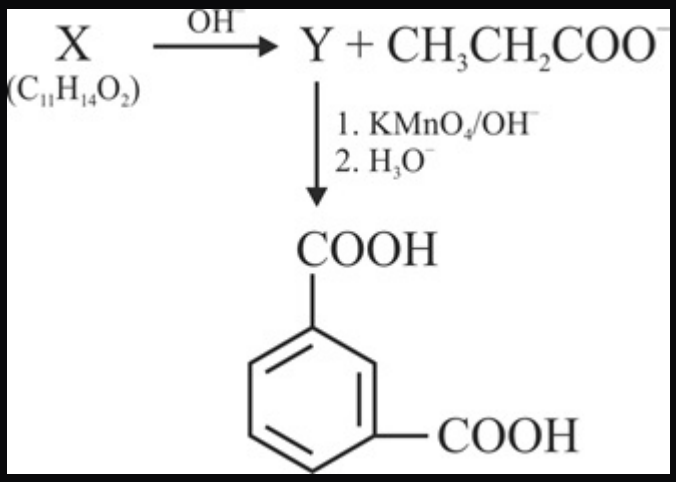
(A) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(B) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(C) If assertion is true but reason is false.

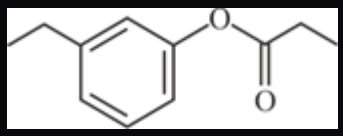
(D) If the assertion and reason both are false.

Q.31

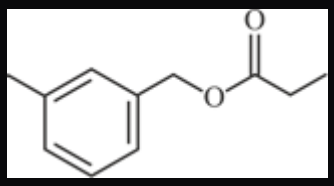


Find the structure of **X**.

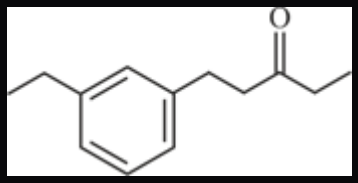
(A)



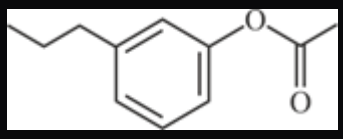
(B)



(C)



(D)



Q.32

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :
Assertion A : The H − O − H bond angle in water molecule is 104. 5°.
Reason R : The lone pair − lone pair repulsion of electrons is higher than the bond pair − bond pair repulsion.

(A) A is false but R is true

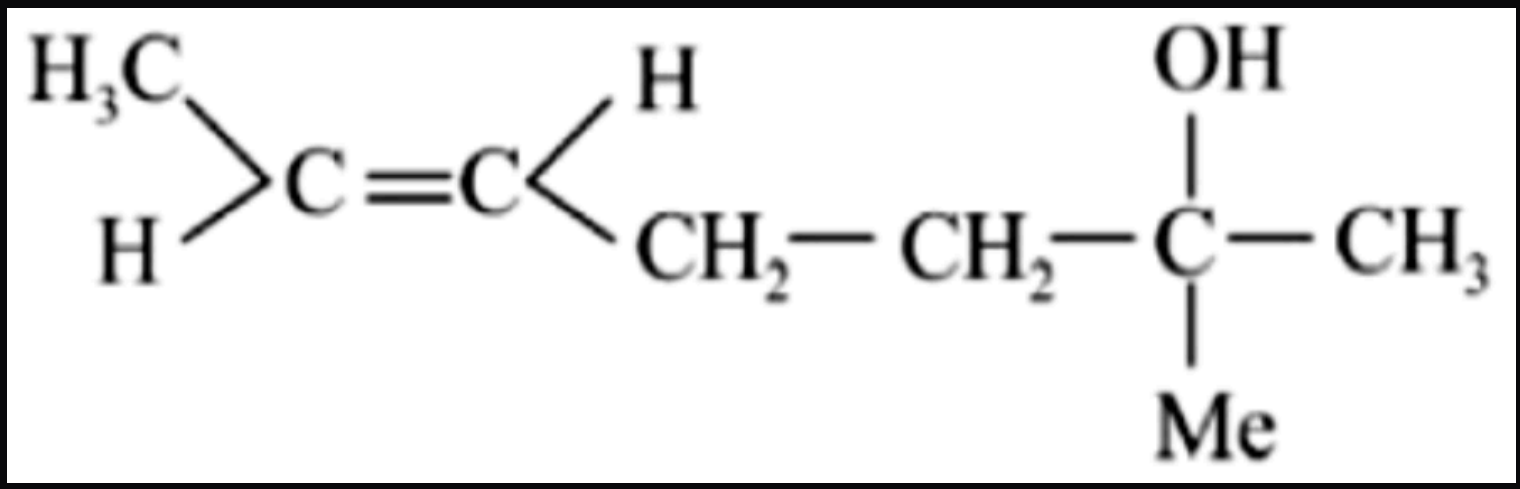
(B) Both A and R are true, but R is not the correct explanation of A

(C) A is true but R is false

(D) Both A and R are true, and R is the correct explanation of A

Q.33

The compound, whose stereo-chemical formula is written below, exhibits x geometrical isomers and y optical isomers.



The value of x and y are :

(A) 4 and 4

(B) 2 and 0

(C) 2 and 2

(D) 4 and 2

Q.34

Consider the following statements.
I. The entropy in isolated system with only $p - V$ work is always maximised at equilibrium.
II. It is possible for the entropy of close system to increase substantially in an irreversible process.
III. Without the help of an external agency, spontaneous process cannot be reversed.
IV. ΔS_{system} is zero for reversible process in any system. Which of the above statement(s) is/are correct?

(A) I, II, III

(B) I, IV

(C) II and III

(D) II, III and IV

Q.35

$Na_2\ B_4O_7 \cdot 10H_2O$ is correctly represented as

(A) $Na_2\ [\ B_4O_5(OH)_4] \cdot 8H_2O$

(B) $2NaBO_2 \cdot Na_2\ B_2O_3 \cdot 1OH_2O$

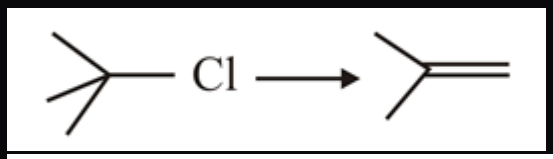
(C) $Na_2\ [\ B_4(H_2O)_4O_7] \cdot 6H_2O$

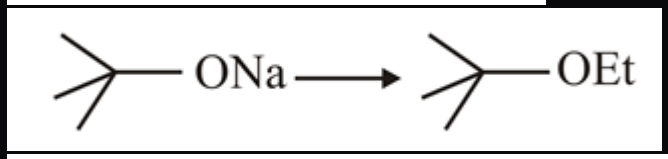
(D) None of the above

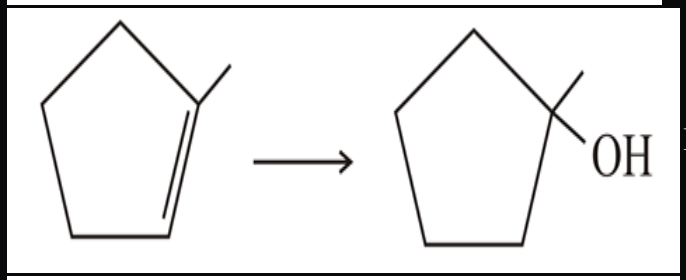
Q.36

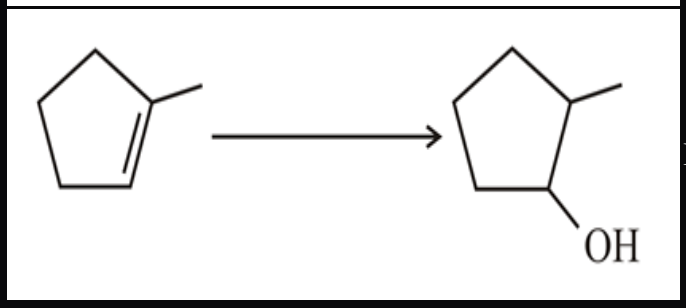
Match the chemical conversions in list I with appropriate reagents in list II and select the correct answer using the code given below the list.

List-I

P. 

Q. 

R. 

S. 

List-II

I. (i) $Hg(OAc)_2$; (ii) $NaBH_4$

II. $NaOEt$

III. $Et - Br$

IV. (i) BH_3; (ii) $H_2O_2/\ NaOH$

(A) $P - II,\ Q - III,\ R - I,\ S - IV$

(B) $P - III,\ Q - II,\ R - I,\ S - IV$

(C) $P - II,\ Q - III,\ R - IV,\ S - I$

(D) $P - III,\ Q - II,\ R - IV,\ S - I$

Q.37

Na^+, Mg^{2+}, Al^{3+} and Si^{4+} ions are isoelectronic. The of ionic radii of these ions.

(A) $Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

(B) $Na^+ < Mg^{2+} < Al^{3+} < Si^{4+}$

(C) $Na^+ > Mg^{2+} > Al^{3+} < Si^{4+}$

(D) $Na^+ < Mg^{2+} > Al^{3+} > Si^{4+}$

Q.38

Identify the incorrect statement:

(A) $CuSO_4$ reacts with KCl in aqueous solution to give Cu_2Cl_2

(B) $CuSO_4$ reacts with KI in aqueous solution to give Cu_2I_2

(C) $CuSO_4$ reacts with $NaOH$ and glucose in aqueous medium to give Cu_2O

(D) $CuSO_4$ on strong heating gives CuO

Q.39

The elements X and Y form compound having molecular formula XY_2 and XY_4 (both are non electrolytes). When dissolved in 20 g benzene, $1 g XY_2$ lowers the freezing point by $2.3^{\circ}C$ whereas 1 g of XY_4 lowers the freezing point by $1.3^{\circ}C$. Molal depression constant for benzene is 5.1. Thus atomic masses of X and Y respectively are

(A) $42.64, 21.10$

(B) $21.10, 42.64$

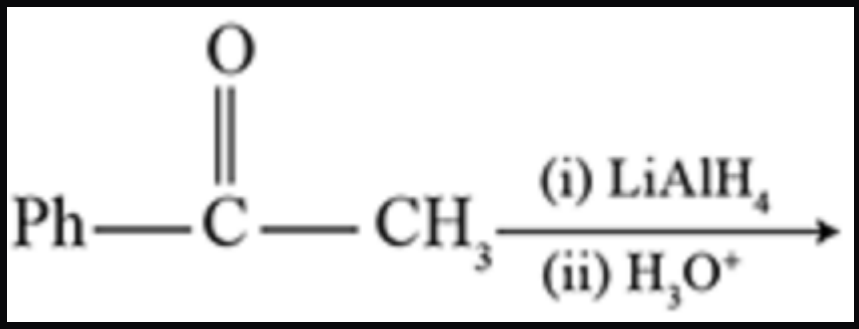
(C) $25.59, 42.64$

(D) $42.64, 25.69$

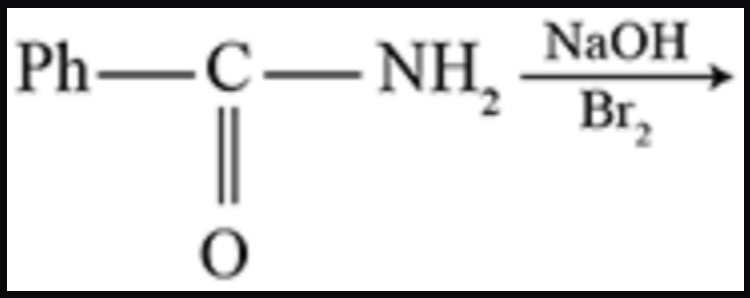
Q.40

In which of the following reaction, cyanide will be obtained as a major product?

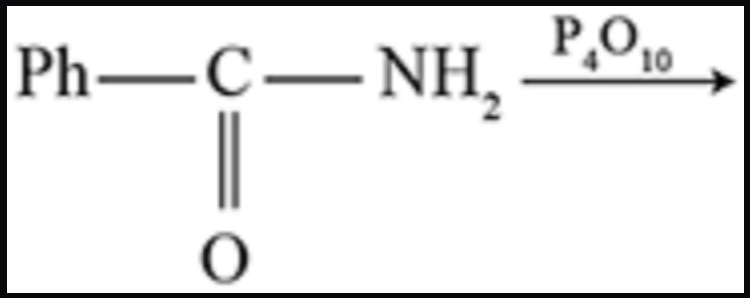
(A)



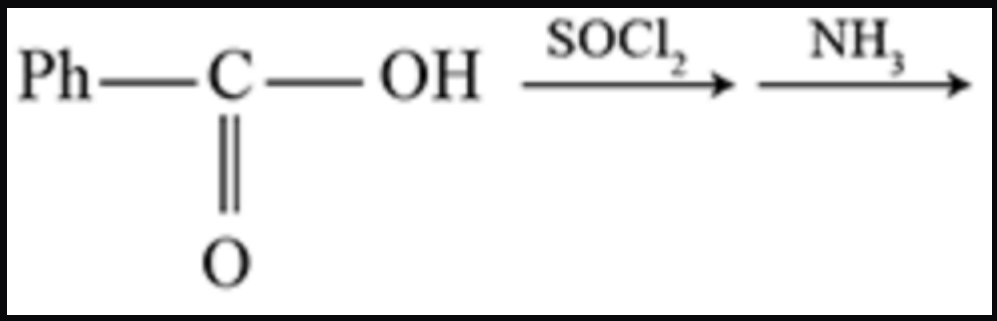
(B)



(C)



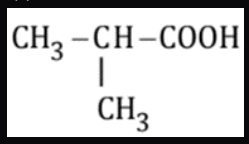
(D)



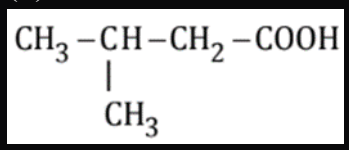
Q.41

Various products formed on oxidation of 2,5-dimethylhexan-3-one are

(i)



(ii)



(iii) CH_3COOH

(iv) HCOOH

(A) (i) and (iii)

(B) (i), (ii) and (iii)

(C) (i), (ii), (iii) and (iv)

(D) (iii) and (iv)

Q.42

The rate of reaction is doubled for every $10^\circ C$ rise in temperature. The increase in rate as result of an increase in temperature from $10^\circ C$ to $100^\circ C$ is how many times of the original rate?

(A) 112

(B) 512

(C) 400

(D) 256

Q.43

Select the correct statement among the following.

(A) Co (III) is stabilised in presence of weak field ligands, while Co (II) is stabilised in presence of strong field ligand.

(B) Four coordinated complexes of Pd (II) and Pt (II) are paramagnetic and square planar.

(C) $[Ni(CN)_4]^{4-}$ ion and $[Ni(CO)_4]$ are tetrahedral and square planar respectively.

(D) Ni^{2+} ion does not form inner orbital octahedral complexes in presence of weak ligand fields.

Q.44

The lanthanide contraction is responsible for the fact that:

(A) Zr and Y have about the same radius

(B) Zr and Nb have similar oxidation state

(C) Zr and Zn have the same oxidation state

(D) Zr and Hf have about the same radius

Q.45

Read the following statements and identify the correct ones regarding polysaccharides.
1. Polysaccharides are made of many sugars joined together.
2. Cellulose is a polysaccharide made of less than 10 monomeric units.
3. Polysaccharides are found as a part of the acid insoluble pellet.
4. In a polysaccharide chain, the right end is called reducing end and the left end is called non-reducing end.

(A) All are correct

(B) All except 2

(C) Only 1 and 4

(D) Only 2

Q.46

Resistance of $0.2M$ solution of an electrolyte is 50Ω. The specific conductance of the solution is $1.3Sm^{-1}$. If resistance of the $0.4M$ solution of the same electrolyte is 260Ω, its molar conductivity is $N \times 10^{-4}Sm^2\ mol^{-1}$, Find $4\ N$.

(A) Submit

Q.47

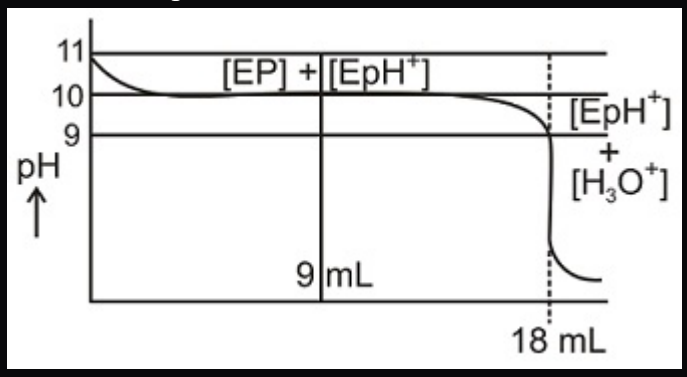
The molarity of HNO_3 in a sample which has density $1.4\ g/mL$ and mass percentage of 63% is _____
(Molecular Weight of $HNO_3 = 63$)

(A) Submit

Q.48

The titration curve for Ephedrine (EP) a weak base dissolved in $182\ mL$ of H_2O when carried out with $1\ M\ HCl$

solution is reproduced below.



At point of equivalence, calculate pH and report the value of $(\mathrm{pH} + \log\ 3)$ as integer answer.

(A) Submit

Q.49

For the equation $\mathrm{CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g),\ K_{1000} = 0.059}$. Exactly 10 g of $\mathrm{CaCO_3}$ is placed in a 10 L container at 1000 K. After equilibrium is reached, what mass of $\mathrm{CaCO_3}$ remains? (Round off to the nearest integer)

(A) Submit

Q.50

What is the number of visible lines when an electron returns from 5th orbit to the ground state in H-spectrum?

(A) Submit

Q.51

The mean and variance of 10 observations are found to be 10 and 5 respectively. On rechecking it is found that an observation 5 is incorrect. If the incorrect observation is replaced by 15, then the correct variance is

(A) 7

(B) 8

(C) 9

(D) 4

Q.52

Let $A = (a_{ij})_{3\times3}$ and $B = (b_{ij})_{3\times3}$, where $b_{ij} = \frac{a_{ij}+a_{ji}}{2}\ \forall i, j$. Number of such matrices A whose elements are selected from the set $\{0, 1,2, 3\}$ such that $A = B$, are

(A) 2^9

(B) 2^{12}

(C) 2^6

(D) 2^8

Q.53

Consider a relation R defined as aRb if $2+ab>0$ where a, b are real numbers. Then, the relation R is

(A) reflexive and symmetric

(B) symmetric and transitive

(C) transitive and reflexive

(D) None of these

Q.54

If $A(3,\ 2,\ 0),\ B(5,\ 3,\ 2)$ and $C(-9,\ 6,\ -3)$ are three points forming a triangle and AD is bisector of the angle $\angle BAC$, then AD meets BC at the point

(A) $\left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$

(B) $\left(-\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$

(C) $\left(\frac{19}{8}, -\frac{57}{16}, \frac{17}{16}\right)$

(D) none of these.

Q.55

The solution of differential equation $(y+x\sqrt{xy}(x+y))dx+(y\sqrt{xy}(x+y)-x)dy=0$, is

(A) $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{2y}}=C$

(B) $\frac{x^2+y^2}{2}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$

(C) $\frac{x^2+y^2}{\sqrt{2}}+2\tan^{-1}\sqrt{\frac{x}{y}}=C$

(D) None of these

Q.56

In an increasing geometric progression, the sum of the first and the last term is 99 , the product of the second and the second last term is 288 and the sum of all the terms is 189 . Then, the number of terms in the progression is equal to

(A) 5

(B) 6

(C) 7

(D) 8

Q.57

The value of $\lim_{x\to\infty}\left[\frac{e^2}{\left(1+\frac{2}{x}\right)^x}\right]^{\frac{x}{2}}$ is equal to

(A) e

(B) e^{-1}

(C) $e^{\frac{1}{2}}$

(D) $e^{-\frac{1}{2}}$

Q.58

If all roots of the equation $f(x) = x^6 - 12x^5 + bx^4 + cx^3 + dx^2 + ex + 64 = 0$ are positive, then which has the greatest numerical (absolute) value

(A) b

(B) c

(C) d

(D) e

Q.59

A bag contains 40 tickets numbered from 1 to 40. Two tickets are drawn from the bag without replacement. The probability that the 2nd ticket is a perfect square given that the 1st ticket was a perfect square is

(A) $\frac{1}{6}$

(B) $\frac{5}{39}$

(C) $\frac{3}{20}$

(D) $\frac{1}{8}$

Q.60

The number of point(s) of minima of the polynomial $y = 10x^6 - 24x^5 + 15x^4 - 40x^2 + 108$ is(are) equal to _____

(A) 1

(B) 2

(C) 3

(D) 4

Q.61

Let $\mathrm{I}_1 = \int_0^1 e^{x^2} dx$ and $\mathrm{I}_2 = \int_0^1 2x^2 e^{x^2} dx$, then the value of $\mathrm{I}_1 + \mathrm{I}_2$ is equal to

(A) 1

(B) 2

(C) e

(D) e^2

Q.62

If principal argument of z_0 satisfying $|z-3| \leq \sqrt{2}$ and $\arg(z-5i) = \frac{-\pi}{4}$ simultaneously is θ, then identify the incorrect statement(s)?

(A) $|z_0| = \sqrt{17}$

(B) $\tan 2\theta = \frac{8}{15}$

(C) $\tan\theta = \frac{-1}{4}$

(D) $|z_0 - 5i| = 4\sqrt{2}$

Q.63

Consider two lines in space as $L_1: \vec{r}_1 = \hat{j} + 2\hat{k} + \lambda\left(3\hat{i} - \hat{j} - \hat{k}\right)$ and $L_2: \vec{r_2} = 4\hat{i} + 3\hat{j} + 6\hat{k} + \mu\left(\hat{i} + 2\hat{k}\right)$ where $\lambda,\ \mu \in R$. If the shortest distance between these lines is $\sqrt{d}$. Then the value of d will be:

(A) 5

(B) 6

(C) 7

(D) 8

Q.64

If a directrix of a hyperbola centered at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e, then:

(A) $4e^4 + 8e^2 - 35 = 0$

(B) $4e^4 - 24e^2 + 35 = 0$

(C) $4e^4 - 24e^2 + 27 = 0$

(D) $4e^4 - 12e^2 - 27 = 0$

Q.65

Coefficients of $x^r [0 \leq r \leq (n-1)]$ in the expansion of $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}(x+2)^2 + \ldots + (x+2)^{n-1}$

(A) ${}^nC_r\left(3^r - 2^n\right)$

(B) ${}^nC_r\left(3^{n-r} - 2^{n-r}\right)$

(C) ${}^nC_r\left(3^r + 2^{n-r}\right)$

(D) None of these

Q.66

If the graph of the function $y = (a-b)^2x^2 + 2(a+b-2c)x + 1(\forall a \neq b)$ is strictly above the x axis, then

(A) $a < b < c$

(B) $a < c < b$

(C) $b < a < c$

(D) $c < b < a$

Q.67

Let $\phi(x,\ y) = 0$ be the equation of a circle if $\phi(0,\ \lambda) = 0$ has equal roots. $\lambda = 2,\ 2$ and $\phi\ (\lambda,\ 0)$ has roots $\lambda\ = \frac{4}{5}, 5$, then, find the centre of the circle.

(A) $\left(2,\ \frac{29}{10}\right)$

(B) $\left(\frac{29}{10},\ 2\right)$

(C) $\left(-2,\ \frac{29}{10}\right)$

(D) None of these

Q.68

The number of solutions of the equation $\sin^{-1}x = (\sin x)^{-1}$ is/are

(A) one

(B) two

(C) three

(D) zero

Q.69

A function $f : Z \to Z$ is defined as $f(n) = \begin{cases} n+1 & n \in \text{ odd integer} \\ \frac{n}{2} & n \in \text{ even integer} \end{cases}$. If $k \in$ odd integer and $f(f(f(k))) = 33$, then the sum of the digits of k is

(A) 7

(B) 5

(C) 9

(D) 8

Q.70

Let $\overrightarrow{u} = \hat{i} + \hat{j}$, $\overrightarrow{v} = \hat{i} - \hat{j}$ and $\overrightarrow{w} = \hat{i} + 2\hat{j} + 3\widehat{k}$. If $\overrightarrow{n}$ is a unit vector such that $\overrightarrow{u}\ .\ \overrightarrow{n} = 0$ and $\overrightarrow{v}\ .\ \overrightarrow{n} = 0$, then $\overrightarrow{w}\ .\ \overrightarrow{n}$ may be equal to

(A) 1

(B) 2

(C) 3

(D) 0

Q.71

The area (in sq. units) enclosed by the solution set of $[x][y] = 3$ is equal to (where $[\cdot]$ denotes greatest integer function)

(A) Submit

Q.72

Given three points on the xy plane $\mathrm{O}(0,0)$, $\mathrm{A}(1,0)$ and $\mathrm{B}(-1,0)$. Point P is moving on the plane satisfying the condition $(\overrightarrow{\mathrm{PA}} \cdot \overrightarrow{\mathrm{PB}}) + 3(\overrightarrow{\mathrm{OA}} \cdot \overrightarrow{\mathrm{OB}}) = 0$. If the maximum and minimum values of $|\overrightarrow{\mathrm{PA}}||\overrightarrow{\mathrm{PB}}|$ are M and m respectively then find the values of $\mathrm{M}^2 + \mathrm{m}^2$.

(A) Submit

Q.73

Let $\tan\alpha, \tan\beta$ and $\tan\gamma; \alpha, \beta, \gamma \neq \frac{(2\mathrm{n}-1)\pi}{2}$ $\mathrm{n} \in \mathrm{N}$ be the slopes of three line segments OA, OB and OC, respectively, where O is origin.If circumcentre of $\Delta\mathrm{ABC}$ coincides with origin and its orthocentre lies on y-axis, then the value of $\left(\frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos\alpha\cos\beta\cos\gamma}\right)^2$ is equal to :

(A) Submit

Q.74

Let $A = \begin{bmatrix} 1 & a^3 & 0 \\ 0 & 1 & b^3 \\ c^3 & 0 & 1 \end{bmatrix}$ where $a, b, c \in R$. If the sum of all non-real roots of the equation $|\mathrm{A} - \mathrm{xI}| = 0$ is $k - mabc, \forall k, m \in Z$, then the value of $k + m$ is equal to

(A) Submit

Q.75

From the string *abacababababcdced*, if 5 letters should be selected, then the number of ways in which this selection can be done is

(A) Submit

Answers & Solutions

Q.1 Answer:

1 cm down

Solution:

Displacement of centre of mass,

$\Delta x_{com} = \frac{m_1\Delta x_1 + m_2\Delta x_2}{m_1+m_2}$

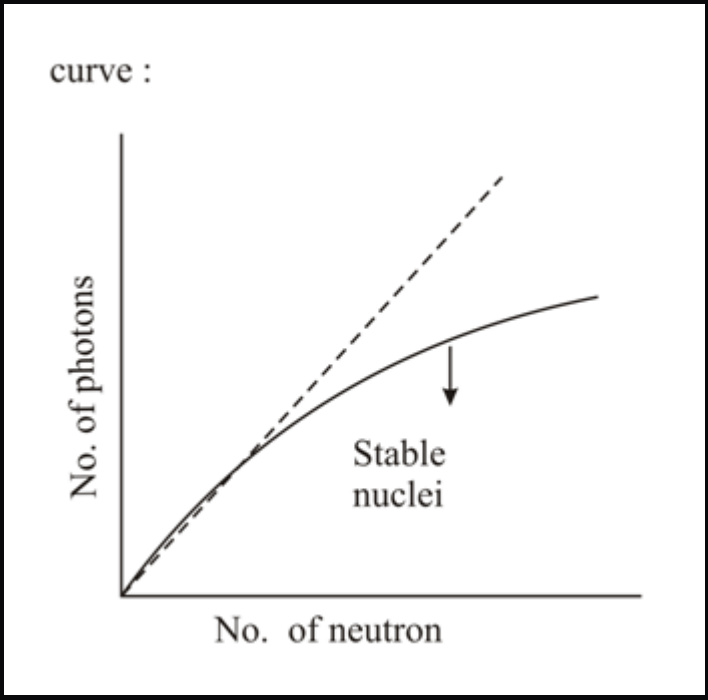
$\Rightarrow 1 = \frac{(10\times 7)+(30x)}{40}$

$\Rightarrow x = -1 \text{ cm}$

Q.2 Answer:

Statement 1 is true, statement 2 is true, statement 2 is correct explanation for the statement 1.

Solution:



So, the plot of atomic number (y-axis) versus the number of neutrons (x-axis) for stable nuclei shows a curvature towards (x-axis) from the line of 45° slope as the atomic number is increased because If the curve does not bend down towards the (x-axis) then the proton-proton repulsion would overcome the attractive force of proton and neutron. Therefore, the curve bends down.

So both the statements are correct and second statement is correct explanation of first statement.

Q.3 Answer:

Material A

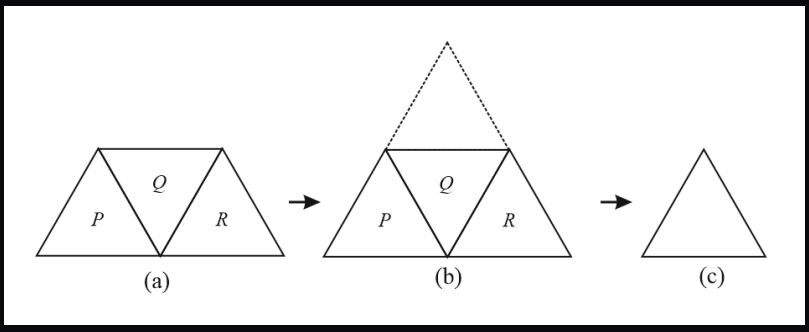
Solution:

Material A is more ductile. It is because, the material A has greater plastic range (portion of graph between the elastic limit and breaking point).

Q.4 Answer:

same deviation.

Solution:

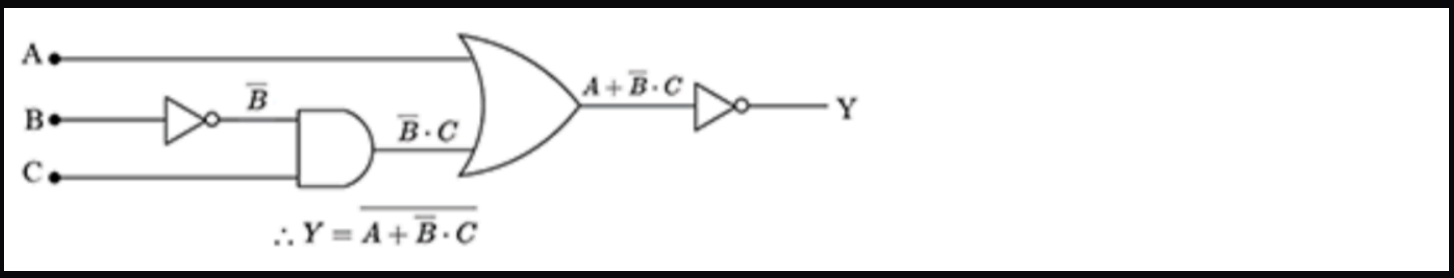


As the P, Q and R have symmetrical geometry and are made up of the same material, so we can assume the combination to be a full prism as shown in figure (b). This will be equivalent to the prism P. So the deviation will be the same deviation as in case of P.

Q.5 Answer:

$\overline{A + \overline{B} \cdot C}$

Solution:



Q.6 Answer:

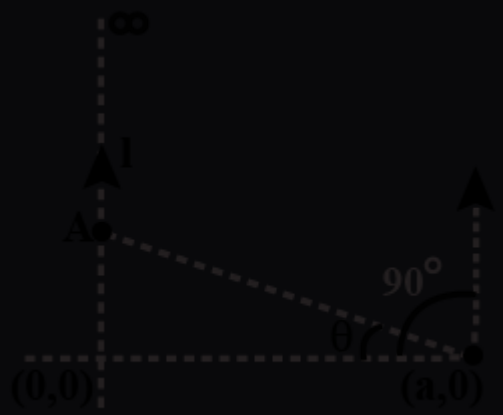
$\frac{\mu_0 i}{4\pi a}\left(1 - \frac{b}{\sqrt{a^2+b^2}}\right)$

Solution:

As shown figure take an element dl at C of wire where, $OC = l$.

Let, $PC = r$ and $\triangle OPC = \phi$.

According to Biot-Savart's law; magnitude of magnetic field induction at P due to current element at C is



$dB = \frac{\mu_0}{4\pi}\frac{idl\sin\theta}{r^2}$

Here, $\theta = 90^\circ + \phi; r = a\sec\phi$

and $\quad l = a\tan\phi;\ \therefore dl = a\sec^2\phi\, d\phi$ (after differentiating)

$\therefore dB = \frac{\mu_0}{4\pi}\frac{i\, a\left(\sec^2\phi\, d\phi\right)\sin(90^\circ+\phi)}{a^2\sec^2\phi}$

$= \frac{\mu_0}{4\pi}\frac{i}{a}\cos\phi\, d\phi$

The magnetic field induction at P is

$$B = \int dB = \int_{90^\circ}^{\phi}\frac{\mu_0}{4\pi}\frac{i}{a}\cos\phi d\phi = \frac{\mu_0}{4\pi}\frac{i}{a}[-\sin\phi)_{90^\circ}^{\phi}$$

$$= \frac{\mu_0}{4\pi}\frac{i}{a}(1-\sin\phi) = \frac{\mu_0 i}{4\pi a}\left(1-\frac{b}{\sqrt{a^2+b^2}}\right)$$

Q.7 Answer:

(a)–(iv), (b) – (ii), (c) – (i), (d) – (iii)

Solution:

According to the EM Wave Spectrum, wavelength range for,

(a) Gamma-rays is fm.

(b) X-rays is A^0

(c) Microwave is in mm

(d) A. M. radio wave is in m

Q.8 Answer:

$-\frac{20}{3}\ \mathrm{ms}^{-2}$

Solution:

Slope of line $= -\frac{2}{3}$

Equation of line is $(v-20) = -\frac{2}{3}(s-0)[$ from,$y = mx] \Rightarrow v = 20 - \frac{2}{3}s \ldots$ (i)

Velocity at $s = 15$ m i.e., $v = \left.\frac{ds}{dt}\right|_{s=15\ \mathrm{m}} = 20 - \frac{2}{3}(15) = 10\ \mathrm{ms}^{-1}$

Differentiating Eq. (i) w.r.t. time,

Acceleration $(a) = \frac{dv}{dt} = -\frac{2}{3}\frac{dS}{dt}$

$\therefore \left.\frac{dv}{dt}\right|_{S=15\ \mathrm{m}} = -\left.\frac{2}{3}\frac{ds}{dt}\right|_{S=15\ \mathrm{m}} = -\frac{20}{3}\ \mathrm{ms}^{-2}$

Q.9 Answer:

$2\pi\sqrt{\frac{3m}{2k}}$

Solution:

For spring connected in parallel $k + k = 2k$

For spring in series $\frac{1}{k_{\mathrm{eq}}} = \frac{1}{k} + \frac{1}{2k} = \frac{3}{2k}$

From the given spring arrangement, $k_{\mathrm{eq}} = \frac{2k}{3}$

As the mass m displaced slightly in the vertical direction it will undergo in SHM.

Time period for SHM is, $T = 2\pi\sqrt{\frac{m}{k_{\mathrm{eq}}}}$

Therefore, the time period of the body of mass m, when displaced slightly in the vertical direction is

$T = 2\pi\sqrt{\frac{3m}{2k}}$

Q.10 Answer:

100 V

Solution:

For series L - C -R circuit

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$
$$= \sqrt{(80)^2 + (40 - 100)^2}$$
$$= 100 \text{ V}$$

Q.11 Answer:

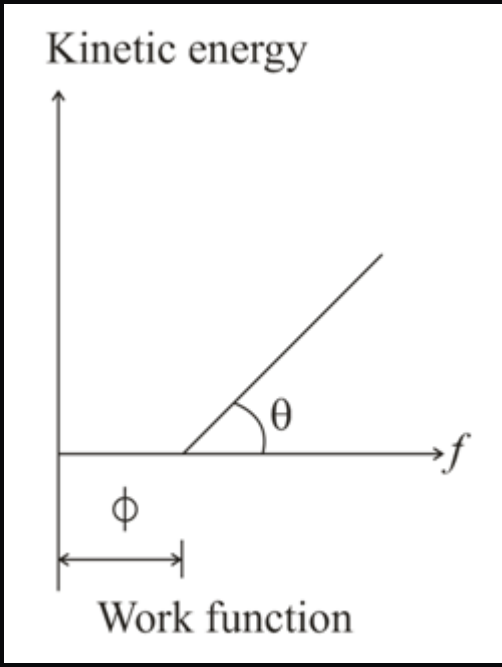
is the same for all metals and independent of the intensity of the radiation

Solution:

According to Einstein's equation,

Kinetic energy = hf - ϕ where kinetic energy and f (frequency) are variables, compare it with equation,

y = mx + c



slope of line = h, where h is Planck's constant.

Hence, the slope is same for all metals and independent of the intensity of radiation.

Q.12 Answer:

$\frac{\pi abi}{2}$, 0

Solution:

Area of ellipse $= \pi ab$

Magnetic dipole moment $= \frac{\pi abi}{2}$

Torque $\vec{\tau} = \overrightarrow{\mathrm{M}} \times \overrightarrow{\mathrm{B}}$

$\tau = MB\sin\theta$

$\because \overrightarrow{\mathrm{M}} || \overrightarrow{\mathrm{B}} \quad \tau = 0$

Q.13 Answer:

$\frac{8}{7}\Omega$

Solution:

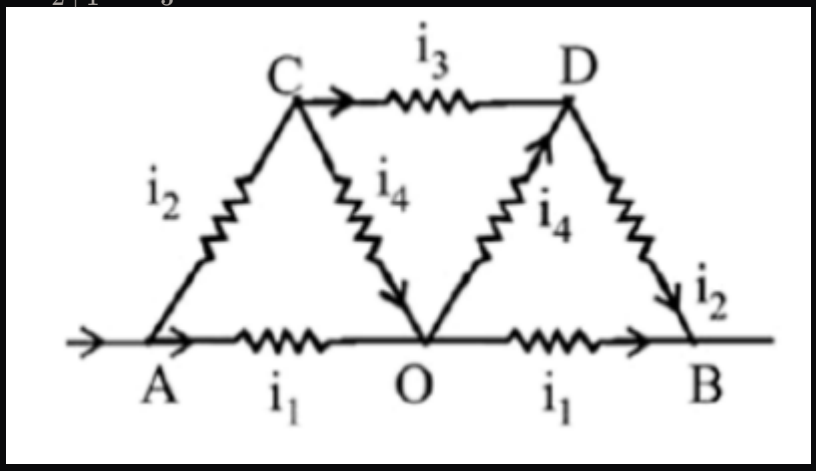
At A current is distributed and at B currents are collected. Between A and B, the distribution is symmetrical. It has been shown in the figure. It appears that current in AO and OB remains same. At O, current i_4 returns back without any change. If we detach O from AB there will not be any change in distribution. Now, CO & OD will be in series hence its total resistance $= 2\Omega$

It is in parallel with CD, so, equivalent resistance

$= \frac{2\times1}{2+1} = \frac{2}{3}\Omega$

This equivalent resistance is in series with $\mathrm{AC\&DB}$ so, total resistance

$= \frac{2\times1}{2+1} = \frac{2}{3}\Omega$



Now $\frac{8}{3}\Omega$ is parallel to AB, that is, 2Ω, so total resistance $= \frac{8/3\times2}{8/3+2} = \frac{16/3}{14/3} = \frac{16}{14} = \frac{8}{7}\Omega$

Q.14 Answer:

Both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

Solution:

According to Maxwell's velocity distribution curve, maximum molecules have same velocity which is called most probable velocity.

Q.15 Answer:

$\mathrm{A} = \mathrm{III}, \mathrm{B} = \mathrm{IV}, \mathrm{C} = \mathrm{I}, \mathrm{D} = \mathrm{II}$

Solution:

Planck's constant, $h = \frac{E\lambda}{c} = \frac{(\mathrm{J})(\mathrm{m})}{\mathrm{m\,s^{-1}}} = \frac{(\mathrm{kg\ m^2\ s^{-2}})(\mathrm{m})}{\mathrm{m\,s^{-1}}} = \mathrm{kg\ m^2\ s^{-1}}$

Torque, $\tau = Fr = \left(\mathrm{kg\ m\,s^{-2}}\right)(\mathrm{m}) = \mathrm{kg\ m^2\ s^{-2}}$

Weight, $W = mg = \mathrm{kg\ m\,s^{-2}}$

Specific volume, $v = \frac{\text{Volume}}{\text{mass}} = \frac{\mathrm{m^3}}{\mathrm{kg}} = \mathrm{kg^{-1}\ m^3}$

Hence, the correct match is $\mathrm{A} = \mathrm{III}, \mathrm{B} = \mathrm{IV}, \mathrm{C} = \mathrm{I}, \mathrm{D} = \mathrm{II}$

Q.16 Answer:

along the diagonal BD

Solution:

The unit positive charge at O. The resultant force due to charge placed at A and C is zero. Force due to charge $(+2q)$ on B will be direction of OD. Force due to charge $(-2q)$ on D will be in direction OD. Therefore, the resultant force will be along the direction OD.

Q.17 Answer:

$\left(\frac{\sqrt{5}}{2}-1\right)R$

Solution:

If g and g_h denote the acceleration due to gravity at the Earth's surface and at the height h above the Earth's surface respectively, then

$$\frac{g_h}{g}=\frac{R^2}{(R+h)^2}$$

But according to the equation

$$g_h=g-\frac{20}{100}g=\frac{80}{100}g=\frac{4}{5}g$$

$$\therefore \frac{4}{5}=\frac{R^2}{(R+h)^2} \text{ or } \sqrt{\frac{4}{5}}=\frac{R}{(R+h)}$$

$$\text{or } \frac{2}{\sqrt{5}}=\frac{R}{R+h} \text{ or } 2R+2h=\sqrt{5}R$$

$$\text{or } 2h=(\sqrt{5}-2)R \text{ or } h=\frac{(\sqrt{5}-2)R}{2}=\left(\frac{\sqrt{5}}{2}-1\right)R$$

Q.18 Answer:

4 N

Solution:

Max force applied on B, till both

move together $\mathrm{F_{max}} = \mu \mathrm{m_A g}\left(1 + \dfrac{\mathrm{m_B}}{\mathrm{m_A}}\right)$

$= 0.5 \times 2 \times 10\left(1 + \dfrac{8}{2}\right)$

$= 10(1 + 4) = 50\ \mathrm{N}$

appplied force, $\mathrm{F} = 20\ \mathrm{N} < \mathrm{F_{max}}$

$\therefore$ both moves together, $a = \dfrac{\mathrm{F}}{\mathrm{m_A} + \mathrm{m_B}}$

$= 20/(2 + 8)$

$= 2\ \mathrm{m/s^2}$

$\therefore$ friction force on $\mathrm{A} = \mathrm{f_A} = \mathrm{m_A a}$

$= 2 \times 2 = 4\ \mathrm{N}$

Q.19 Answer:

-2

Solution:

As is known,

$$\mathrm{PE} = -2\mathrm{KE}$$

i.e., $E_p = -2E_k$

or $\frac{E_p}{E_k} = -2$

Q.20 Answer:

$\frac{4}{5}\frac{P_0 V_0}{R}$

Solution:

$P = P_0\left[1 + \left(\frac{2v_0}{v}\right)^2\right]^{-1}$

if $v = v_0$ **and if** $v = 2v_0$

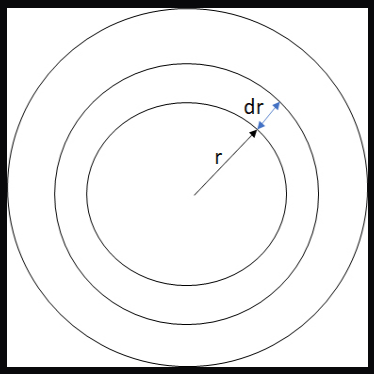
$\Rightarrow P = \frac{P_0}{5}$ **and** $P = \frac{P_0}{2}, T = \frac{Pv}{nR}$

$\Delta T = T_f - T_i = \frac{\frac{P_0}{2} \times 2v_0}{nR} - \frac{\frac{P_0}{5} \times v_0}{nR} = \frac{4}{5}\frac{P_0 v_0}{nR}$

Q.21 Answer:

—

Solution:



The moment of inertia of a hollow sphere shell in terms of mass and radius of the shell is, $I_0 = \frac{2}{3}mr^2$ **. Here, for sphere** A**:**

$I_A = \int_0^R \frac{2}{3}r^2 \cdot 4\pi r^2 dr K\frac{r}{R} \Rightarrow I_A = \frac{8\pi K}{3R}\int_0^R r^5 dr = \frac{8\pi K}{3R}\frac{R^6}{6}$.

Also, for sphere B**:**

$I_B = \int_0^R \frac{2}{3}r^2 \cdot 4\pi r^2 dr \frac{K}{R^5} \cdot r^5 = \frac{8\pi K}{3R^5} \cdot \frac{R^{10}}{10}$

The ratio, $\frac{I_B}{I_A} = \frac{6}{10}$.

"

Q.22 Answer:

—

Solution:

$F = -\frac{dv}{dx} = +3x^2 - 12x = 0$

$(x-4)x = 0$

$7x = 0, 4$

$x = 0, U = 15\text{ J}, KE = \frac{1}{2}(2)(16 \times 5) = 80\text{ J}$

$x = 4, U = -4^3 + 6.4^2 + 15$

$= 47\text{ J} \Rightarrow KE = (15 + 80) - 47$

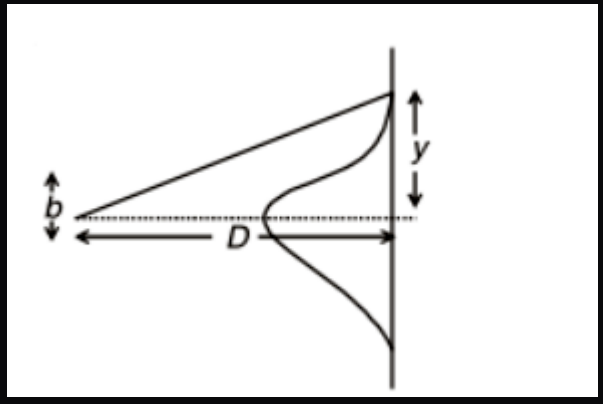
$\frac{1}{2}(2)v^2 = 48\text{ J}$

$v^2 = 48\text{ J}$

Q.23 Answer:

—

Solution:



$b\sin\theta = 1\lambda$

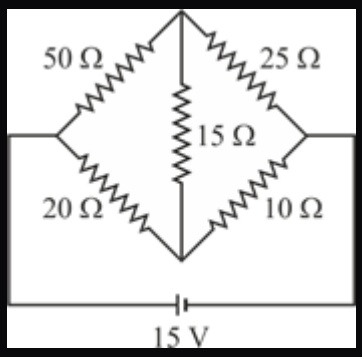
$\sin\theta = \frac{\lambda}{b}$
$\sin\theta \approx \tan\theta = \frac{\lambda}{b} = \frac{y}{D} \Rightarrow 2y = \frac{2D\lambda}{b}$
$= 2.4 \text{ mm}$

Q.24 Answer:

—

Solution:

Given, arrangement is a balanced Wheatstone bridge clearly, $\frac{50}{20} = \frac{25}{10}$

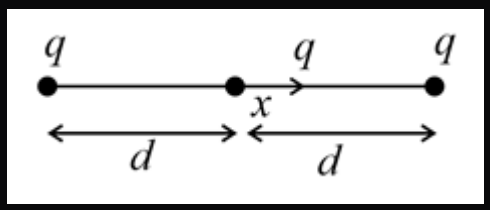


So, 15Ω **resistance can be removed so equivalent resistance is**
$R_{\text{eq}} = \frac{75\times30}{75+30} = \frac{150}{7}$
So, current, $I = \frac{15}{150} \times 7 = 0.7 \text{ A}$

Q.25 Answer:

—

Solution:

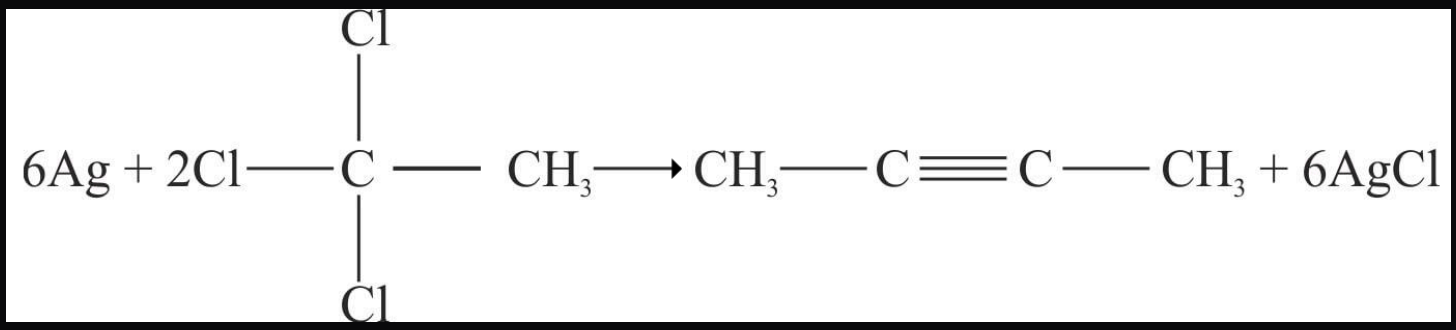


Net force on free charged particle,
$F = \frac{kq^2}{(d+x)^2} - \frac{kq^2}{(d-x)^2}$
$F = -kq^2\left[\frac{4dx}{(d^2-x^2)^2}\right]$
Since $d >>> x$**, so neglecting** x **in denominator.**
$a = -\frac{4kq^2d}{m}\left(\frac{x}{d^4}\right)$
$a = -\left(\frac{4kq^2}{md^3}\right)x$
So, angular frequency
$\omega = \sqrt{\frac{4\,kq^2}{md^3}}$
$\omega = \sqrt{\frac{4\times9\times10^9\times10}{1\times10^{-6}\times1^3}}$
$\omega = 6\times10^8 \text{ rad sec}^{-1}$

Q.26 Answer:

I, II & III

Solution:



$$HC \equiv CH + Na \longrightarrow NaC \equiv CNa \xrightarrow{CH_3Cl} CH_3 - C \equiv C - CH_3$$

Q.27 Answer:

$2.303 \times RT - 2.2F$

Solution:

$\Delta G = \Delta G^{\circ} + 2.303 \times RT \times \log Q,$

$\Delta G = -nFE^{\circ} + 2.303 \times RT \times \log Q.$

Given, $E^{\circ} = 1.1$ V **and** $n = 2,$

$\Delta G = (-2 \times 1.1 \times F) + 2.303 \times RT \times \log \frac{[Zn^{+2}]}{[Cu^{+2}]},$

$\Delta G = -2.2F + 2.303 \times RT.$

Q.28 Answer:

$C_6H_{10}O_4$

Solution:

Misplaced \hline

&

Simplest ratio of atoms

\ \ \hline \mathrm{C} & 49.3 & 49.3 / 12=4.1 & 4.1 / 2.74=1.5 \ \ & & 1.5 \times 2=3 & \ \ \mathrm{H} & 6.84 & 6.84 / 1=6.84 & 6.84 / 2.74=2.5 \ \ & & & =2.5 \times 2=5 \ \ \mathrm{O} & 43.86 & 43.86 / 16=2.74 & 2.74 / 2.74=1 \ \ & & & 1 \times 2=2 \ \ \hline \ \end{array}

$\therefore$ **Empirical formula** $= C_3H_5O_2$

Empirical formula mass

$= (3 \times 12) + (5 \times 1) + (2 \times 16) = 36 + 5 + 32 = 73$

Molecular mass $= 2\times$ **Vapour density**

$= 2 \times 73 = 146$

$n = \frac{\text{molecular mass}}{\text{empirical formula mass}} = 146/73 = 2$

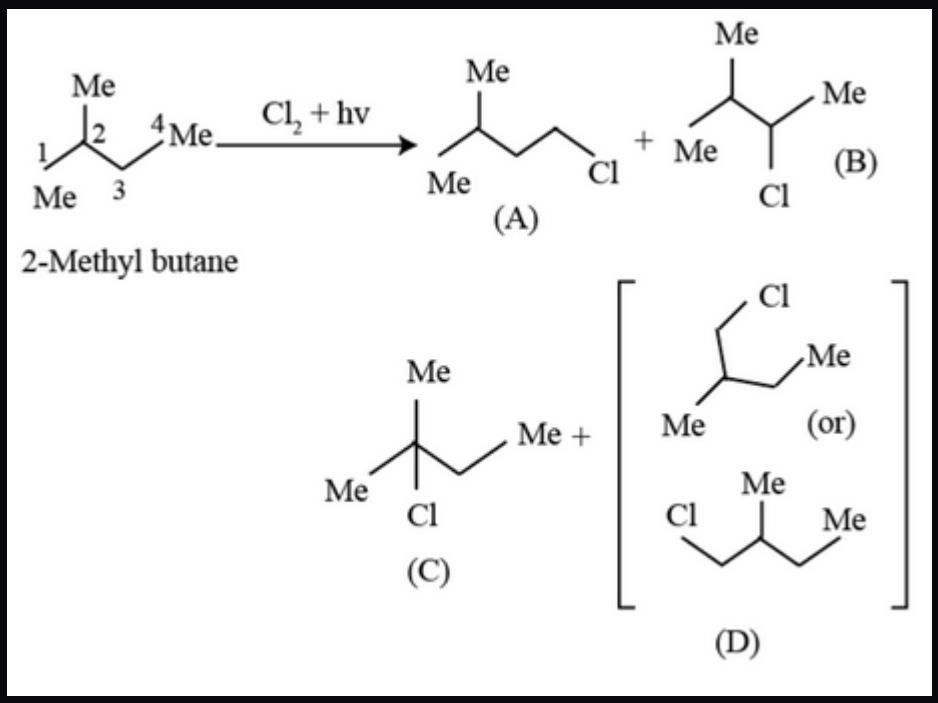
Molecular formula $=$ **Empirical formula** $\times 2$

$= (C_3H_5O_2) \times 2 = C_6H_{10}O_4$

Q.29 Answer:

$14\%, 35\%, 23\%, 28\%$

Solution:



There are four different $(\mathrm{A}, \mathrm{B}, \mathrm{C}$ and $\mathrm{D})$ monochloro products.

			Relative amount
(A)	Me, Me, Cl	(A) is obtained from the reaction of three equivalent 1° H atoms	$3 \times 1 = 3.0$
(B)	Me, Me, Cl	(B) is obtained from the reaction of two equivalent 2° H atoms	$3.82 \times 1 = 7.6$
(C)	Me, Me, Cl	(C) is obtained from the reaction of one equivalent 3° H atoms	$1 \times 5 = 5.0$
(D)	Cl, Me, Me, or; Me, Me, Cl	(D) is obtained from the reaction of six equivalent 1° H atoms	$6 \times 1 = 6.0$ Total =21.6

Therefore, percentages of the A, B, C and D monochloro products are as follows:

$(A) = \frac{3}{21.6} \times 100 = 13.9 \approx 14\%$

$(B) = \frac{7.6}{21.6} \times 100 = 35.18 \approx 35\%$

$(\mathrm{C}) = \frac{5.0}{21.6} \times 100 = 23.15 \approx 23\%$

$(D) = \frac{6.0}{21.6} \times 100 = 27.8 \approx 28\%$

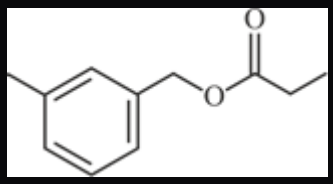
Q.30 Answer:

If both assertion and reason are true but reason is not the correct explanation of the assertion.

Solution:

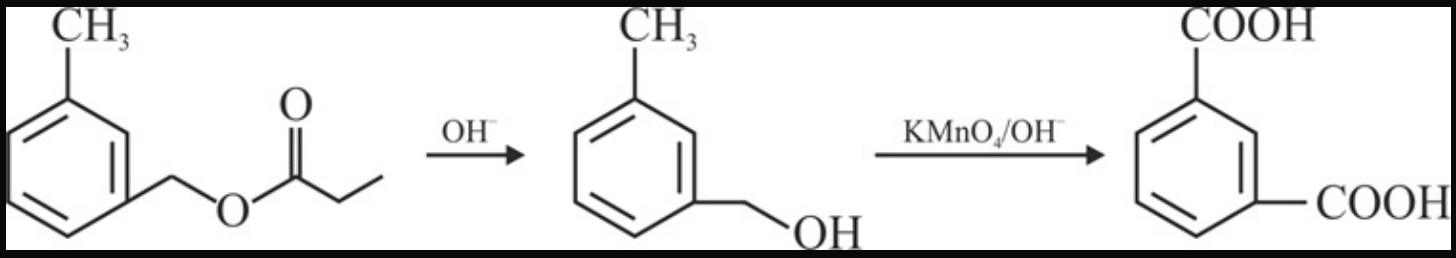
Chromatography is used to separate almost any given mixture. Whether coloured or colourless into its constituents and to test the purites of these constituents.

Q.31 Answer:



Solution:

1) First reaction is hydrolysis of ester in basic medium. Ester on hydrolysis gives Carboxylic acid and Alcohol,



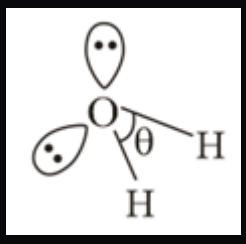
2) Second reaction is side chain Oxidation of Benzene in presence of $\mathrm{KMnO_4}$ to give Carboxylic acid group.

Q.32 Answer:

Both A and R are true, and R is the correct explanation of A

Solution:

$\mathrm{H_2O}$



$\theta = 104.5^{\circ}$

The hybridisation of oxygen is water molecule is $\mathrm{sp^3}$.

So electron geometry of water molecule is tetrahedral, and the bond angle should be $109^{\circ}28''$ but as we know that lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion because lone pair occupies more space around central atom than that of bond pair, The $\mathrm{H-O-H}$ bond angle in water molecule is 104.5°.

Q.33 Answer:

2 and 0

Solution:

$X =$ cis and trans

$Y = 0$, due to absence of chirality

Q.34 Answer:

I, II, III

Solution:

Statement (I) - The entropy isolated system with $P - V$ work only is always a maximized at equilibrium.

Statement (II) - It is possible for the entropy of a close system to decrease substantially in air irreversible process.

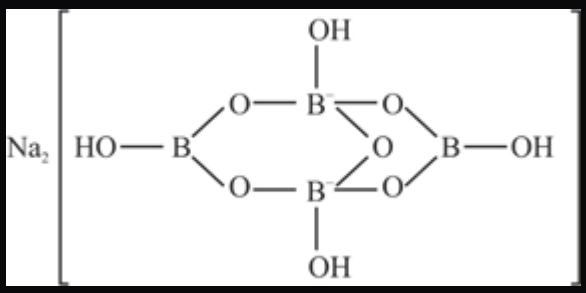
Statement IV - delta S_{system} is zero for reversible process in an isolated system.

Q.35 Answer:

$Na_2\,[\,B_4O_5(OH)_4] \cdot 8H_2O$

Solution:

Borax molecule is actually made up of two tetrahedral and two triangular units joined as follows:



So, correct representation of borax is,

$Na_2\,[\,B_4O_5(OH)_4] \cdot 8H_2O$

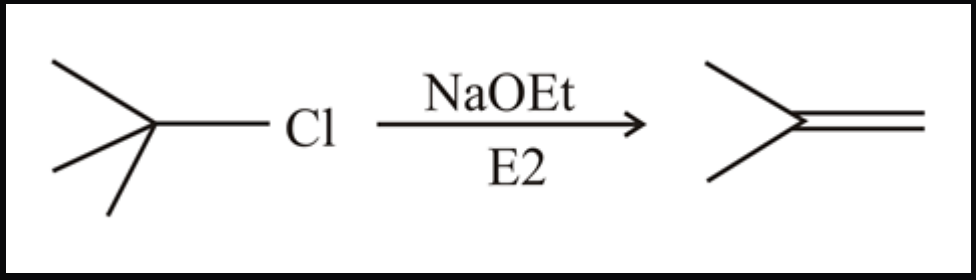
Q.36 Answer:

P – II, Q – III, R – I, S – IV

Solution:

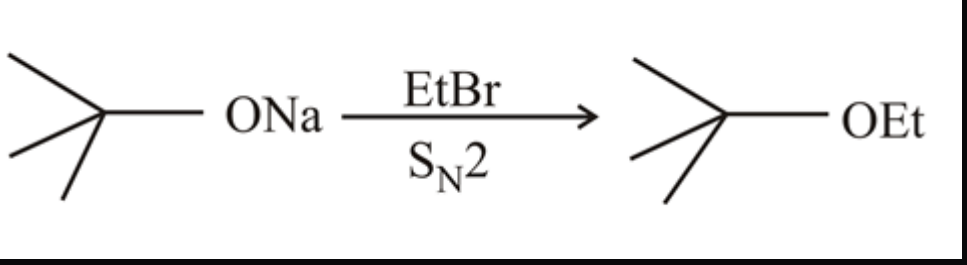
P: NaOEt

P is dehydrohalogenation of haloalkanes to give alkenes in the presence of a strong base like NaOEt.



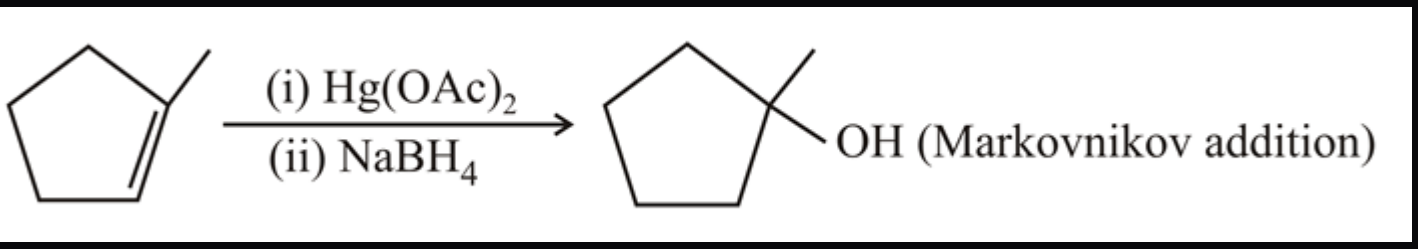
Q: EtBr

Q is Williamson ether synthesis which is an organic reaction, forming an ether from an organohalide (EtBr) and an alkoxide. Thus, Q matches with EtBr which is an organohalide.



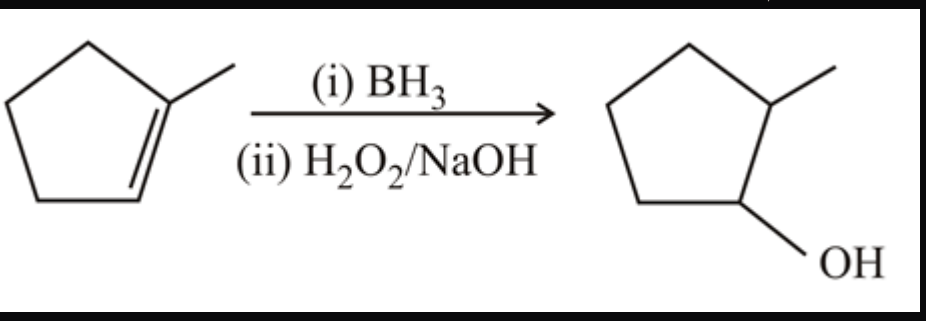
R: $Hg(OAc)_2$; $NaBH_4$

In R, the reagent used to produce alcohol from alkene by Markovnikov's rule is $Hg(OAc)_2$; $NaBH_4$



S: BH_3; H_2O_2/ NaOH

S is hydroboration oxidation which is an anti-Markovnikov's reaction, which converts an alkene into alcohol. The process results in the syn-addition of hydrogen and anti-addition of a hydroxyl group, where the double bond has been. The reagent used is BH_3; H_2O_2/ NaOH.



Q.37 Answer:

$Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

Solution:

Species	Na^+	Mg^{2+}	Al^{3+}	Si^{4+}
Protons	11	12	13	14
Electrons	10	10	10	10

Size of isoelectronic cations decreases with increase in magnitude of nuclear charge

∴ Order of decreasing size is

$Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$

Q.38 Answer:

$CuSO_4$ reacts with KCl in aqueous solution to give Cu_2Cl_2

Solution:

$CuSO_4$ **reacts with** KI **to give** Cu_2I_2 **(white ppt.) but not with** KCl.

Fehling solution mainly contains $CuSO_4$ **and** $NaOH$ **When warmed with glucose (with** CHO **group) gives red ppt. of** Cu_2O

$CuSO_4$ **on heating gives** CuO

Q.39 Answer:

25.59, 42.64

Solution:

$$\Delta T_f = K_f \frac{w}{M \cdot W} \times 1000$$
$$2.3 = 5.1 \times \frac{1}{M_{XY_2}} \times \frac{1000}{20}$$
$$M_{XY_2} = 110.87$$

and $1.3 = 5 \times \frac{1}{M_{XY_4}} \times \frac{1000}{20}$

$M_{XY_4} = 196.15$

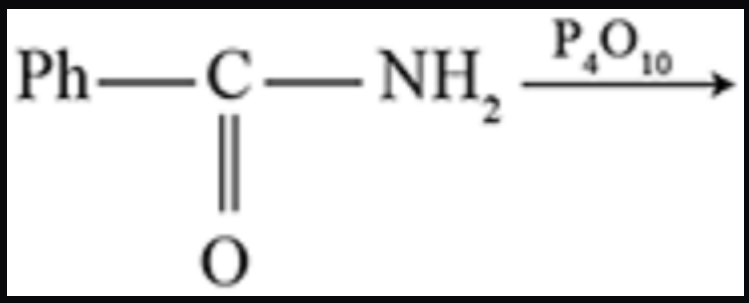
Now, let atomic mass of $X = a$ and atomic mass of $Y = b$ Thus, $a + 2b = 110.87$

and $\frac{-a \pm 4b = -196.15}{-2b = -85.28}$

$b = 42.64$

and thus $a = 25.59$

Q.40 Answer:



Solution:

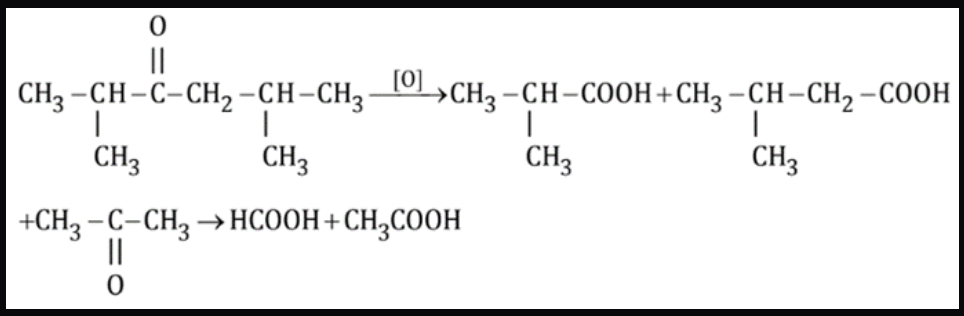
$$Ph - \underset{\underset{O}{\|}}{C} - NH_2 \xrightarrow[-H_2O]{P_4O_{10}} Ph - C \equiv N$$

P_4O_{10} is a strong dehydrating agent

Q.41 Answer:

(i), (ii), (iii) and (iv)

Solution:



Q.42 Answer:

512

Solution:

$\frac{k_{t+10}}{k_t} = 2$

The increase in rate as result of increase in temperature from $10^{\circ}C$ to $100^{\circ}C$ is $r_n = (2)^9 r_1$

Q.43 Answer:

Ni^{2+} ion does not form inner orbital octahedral complexes in presence of weak ligand fields.

Solution:

This problem includes conceptual mixing of CFSE, CFT and magnetism in coordination compounds. This problem can be solved by usual understanding of each statement as explained below:

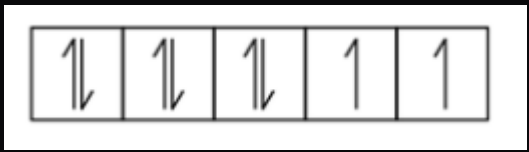
(I) Both Co^{2+} and Co^{3+} are stabilised in presence of strong of field ligands due to high CFSE values.

(II) Due to higher Zeft values on the valence shell Pd^{2+} and Pt^{2+} cations, they always form inner orbital complexes but are diamagnetic in nature and square planar.

(III) $[Ni(CO)_4]$; sp^3, tetrahedral, $\mu = 0BM$

$[Ni(CN)_4]^{4-}$; dsp^2, square planar, $\mu = 0BM$

(IV) In Ni^{+2}, pairing of electron do not occur in presence of weak field ligands, so they form an outer orbital complex.



Electronic distribution unaffected.

Q.44 Answer:

Zr and Hf have about the same radius

Solution:

Zr and Hf belong to 4*d* & 5*d* series in same group have same atomic size due to Lanthanide Contraction. The major cause for lanthanide contraction is due to the inappropriate shielding of the 4f electrons due to the improper shape of the f-orbitals. As the atomic number increases in the lanthanide series, for every proton in the nucleus the extra electron goes to fill the 4f-orbitals.

Q.45 Answer:

All except 2

Solution:

Polysaccharides (type of carbohydrates) comes under the class of macromolecules. Macromolecules form the part of retentate (acid insoluble fraction) when a living tissue is treated with trichloroacetic acid. Polysaccharides form long chains of sugars made by joining same or different monosaccharides together. Cellulose is a polysaccharide having more than 10 units of only one type of monosaccharide - glucose. In a polysaccharide chain (say glycogen), the right end is called the reducing end and the left end is called the non-reducing end as the right end cannot reduce oxidizing agents. Thus, only statement 2 is incorrect.

Q.46 Answer:

—

Solution:

Specific conductance = conductance × cell constant

$1.3 = \frac{1}{50} \times$ **cell constant**

$\therefore$ **Cell constant** $= 1.3 \times 50\ \mathrm{m}^{-1}$

$= 65\ \mathrm{m}^{-1} = \left(\frac{65}{100}\right) \mathrm{cm}^{-1}$

Molar conductivity

$= \frac{1000 \times \text{ conductance} \times \text{ cell constant}}{\text{molarity}} = \frac{1000}{0.4} \times \frac{1}{260} \times \frac{65}{100} = 6.25\ \mathrm{S\ cm^2\ mol^{-1}}$

$= 6.25 \times 10^{-4} \mathrm{Sm^2\ mol^{-1}}$

$N = 6.25, 4N = 25$

Q.47 Answer:

—

Solution:

$63\% \frac{w}{w} \longrightarrow HNO_3$ **solution**

$M = \frac{63 \times 1.4}{63 \times 100} \times 1000 mole/L$

$M = 14 mole/L$

Q.48 Answer:

—

Solution:

At midpoint of titration $[\mathrm{E_P}] = [\mathrm{E_PH^+}]$

$\mathrm{pOH} = \mathrm{pK_b} + \log \frac{[\mathrm{E_PH^+}]}{[\mathrm{E_P}]} \Rightarrow 4 = \mathrm{pK_b} \therefore \mathrm{K_b} = 10^{-4}$

For conjugate pair of acid and base $\mathrm{K_a} \cdot \mathrm{K_b} = 10^{-14}$

$\mathrm{K_a} = 10^{-10}\ \mathrm{M}$

At point of equivalence mM **of** HCl **added**$= 18\ \mathrm{mM}.$

$$\begin{array}{ccccc} \mathrm{E_P} & + & \mathrm{H^+} & \rightarrow & \mathrm{E_PH^+} \\ 18\ \mathrm{mM} & & 18\ \mathrm{mM} & & 0 \\ -18\ \mathrm{mM} & & -18\ \mathrm{mM} & & +18\ \mathrm{mM} \end{array}$$

$[\mathrm{E_PH^+}]$ **at equivalence point** $= \frac{18\ \mathrm{mM}}{(182+18)\mathrm{mL}} = 9 \times 10^{-2}\ \mathrm{M}$

$\left[\mathrm{H_3O^+}\right] = \sqrt{\mathrm{K_a} \cdot \mathrm{C}} = \sqrt{10^{-10} \times 9 \times 10^{-2}} = 3 \times 10^{-6}\ \mathrm{M}$

$\mathrm{pH} = 6 - \log 3$

$\mathrm{pH} + \log 3 = 6 - \log 3 + \log 3 = 6.$

Q.49 Answer:

—

Solution:

Given equilibrium reaction: $\mathrm{CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g)}$

Molar mass of $\mathrm{CaCO_3}\ =\ 100\ \mathrm{g}$

Number of mole of $\mathrm{CaCO_3}\ =\ \frac{10}{100} = 0.1\ \mathrm{mole}$

Concentration of $\mathrm{CaCO_3}$ $= 0.01 \frac{\text{mole}}{\text{L}}$

Initial concentration: $\underset{0.01}{\mathrm{CaCO_3}}\left(\mathrm{s}\right) \rightarrow \underset{0}{\mathrm{CaO}}\left(\mathrm{s}\right) + \underset{0}{\mathrm{CO_2}}\left(\mathrm{g}\right)$

Equilibrium concentration: $\underset{0.01-\mathrm{a}}{\mathrm{CaCO_3}}\left(\mathrm{s}\right) \rightarrow \underset{\mathrm{a}}{\mathrm{CaO}}\left(\mathrm{s}\right) + \underset{\mathrm{a}}{\mathrm{CO_2}}\left(\mathrm{g}\right)$

$\mathrm{CO_2}$ does not participate in equilibrium reaction.

The equilibrium constant for given reaction at $1000\ \mathrm{K}$ is given as $\mathrm{K} = \frac{[\mathrm{CaO}]}{[\mathrm{CaCO_3}]}$.

Given, $\mathrm{K}_{1000} = 0.059$

$0.059 = \frac{\mathrm{a}}{0.01-\mathrm{a}} 0.00059 = 1.059\ \mathrm{a} \ \Rightarrow\ \mathrm{a} = \frac{0.00059}{1.059} \mathrm{a} = 0.00056.$

Hence, concentration of $\mathrm{CaCO_3} = 0.01 - 0.00056 = 0.00939\ \frac{\text{mole}}{\text{L}}$.

Mole of $\mathrm{CaCO_3}$ remain $= 0.00939 \times 10 = 0.0939\ \text{mole}.$

Hence, Mass of $\mathrm{CaCO_3}$ remaining $= 0.0939 \times 100\ =\ 9.39\ \mathrm{g}.$

Q.50 Answer:

—

Solution:

Lines of Balmer series obtained in visible region.

When an electron comes from any higher orbit to second orbit, these lines are known as Balmer series $(\mathrm{n_1} = 2,\ \mathrm{n_2}\ > \mathrm{n_1})$.

An electron can return from fifth orbit to second orbit in different styles and steps as follows;

$5 \rightarrow 2,$

$4 \rightarrow 2,$

$3 \rightarrow 2$

Here 3 lines occur in visible region.

Q.51 Answer:

4

Solution:

$\bar{x}_{\text{old}} = 10 = \frac{\Sigma x_i}{10} \Rightarrow \Sigma x_{i\ \text{old}} = 100$

$\Sigma x_{\text{inew}} = 100 - 5 + 15 = 110$

$\bar{x}_{\text{new}} = \frac{110}{10} = 11$

$\mathrm{Var}_{\text{old}} = 5 = \frac{\Sigma x_{\text{iold}}^2}{10} - \left(\bar{x}_{\text{old}}\right)^2$

$\Sigma x_{\text{iold}}^2 = 1050$

$\Sigma x_{\text{inew}}^2 = 1050 - 25 + 225$

$= 1250$

Var $_{\text{new}} = \frac{1250}{10} - (11)^2 = 125 - 121 = 4$

Q.52 Answer:

2^{12}

Solution:

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ **and** $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$

$\because A = B \Rightarrow a_{ij} = b_{ij} = \frac{a_{ij}+a_{ji}}{2}$

$\Rightarrow a_{ij} = a_{ji}$

$\Rightarrow$ **A is a symmetric matrix**

Hence, the number of such matrix $= 4 \times 4 \times 4 \times 4 \times 4 \times 4 = 4^6 = 2^{12}$

Q.53 Answer:

reflexive and symmetric

Solution:

Relation R **is reflexive, since** $2 + a \times a > 0, \forall$ **real number** a**.**

It is symmetric, since $2 + ab > 0 \Rightarrow 2 + ba > 0$**.**

However, R **is not transitive.**

Consider three real numbers $2, -\frac{1}{6}$ **and** -2 **. We have**

$2 + 2 \times \left(-\frac{1}{6}\right) = \frac{5}{3} > 0$

and $2 + \left(-\frac{1}{6}\right)(-2) = \frac{7}{3} > 0$

So, $2R\left(-\frac{1}{6}\right)$ **and** $\left(-\frac{1}{6}\right)R(-2)$

But, $2 \not{R} (-2)$ **since** $2 + 2(-2) = -2 \not> 0$

Q.54 Answer:

$\left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$

Solution:

The bisector of $\angle BAC$**, i.e.,** AD **divides the side** BC **in the ratio** $AB : BC$**, i.e.,** $\frac{BD}{CD} = \frac{AB}{AC}$

Now, $AB = \sqrt{(3-5)^2 + (2-3)^2 + (0-2)^2}$

$= \sqrt{4+1+4} = 3$

$AC = \sqrt{(-9-3)^3 + (6-2)^2 + (-3-0)^2}$

$= \sqrt{144+16+9} = 13$

Let co-ordinates of $D(x,\ y,\ z)$

$x = \frac{3\times(-9)+13\times5}{3+13} = \frac{38}{16} = \frac{19}{8}$

$y = \frac{3\times6+3\times13}{3+13} = \frac{57}{16}$

$z = \frac{-3\times3+13\times2}{3+13} = \frac{17}{16}$

$\therefore\ D\left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$

Q.55 Answer:

$\frac{x^2+y^2}{2} + 2\tan^{-1}\sqrt{\frac{x}{y}} = C$

Solution:

The given equation can be written as

$(ydx - xdy) + x\sqrt{xy}(x+y)dx + y\sqrt{xy}(x+y)dy = 0$

$\Rightarrow \quad (ydx - xdy) + (x+y)\sqrt{xy}(xdx + ydy) = 0$

$\Rightarrow \quad \frac{ydx-xdy}{y^2} + \left(\frac{x}{y}+1\right) \cdot \sqrt{\frac{x}{y}} d\left(\frac{x^2+y^2}{2}\right) = 0$

$\Rightarrow \quad d\left(\frac{x}{y}\right) + d\left(\frac{x^2+y^2}{2}\right)\left(\frac{x}{y}+1\right)\sqrt{\frac{x}{y}} = 0$

$\Rightarrow \quad d\left(\frac{x^2+y^2}{2}\right) + \frac{d\left(\frac{x}{y}\right)}{\left(\frac{x}{y}+1\right)\cdot\sqrt{\frac{x}{y}}} = 0$

Integrating both the sides, we get

$\Rightarrow \quad \frac{x^2+y^2}{2} + 2\tan^{-1}\sqrt{\frac{x}{y}} = C$

Q.56 Answer:

6

Solution:

G.P. is increasing, i.e. $r > 1$ **Given,** $a + ar^{n-1} = 99, ar \cdot ar^{n-2} = 288$ **and** $\frac{a(1-r^n)}{1-r} = 189$

$a\left(1 + r^{n-1}\right) = 99$ **and** $a^2 r^{n-1} = 288$

$\Rightarrow a\left(1 + \frac{288}{a^2}\right) = 99$

$\Rightarrow a^2 + 288 = 99a \Rightarrow a^2 - 99a + 288 = 0$

$\Rightarrow a = 3, 96 \Rightarrow r^{n-1} = \frac{288}{a^2} = 32, \frac{1}{32}$

As $r > 1 \Rightarrow r^{n-1} = 32$ **for** $a = 3$

Now, $\frac{a(1-r^n)}{1-r} = 189 \Rightarrow \frac{3\left(1-r\cdot r^{n-1}\right)}{1-r} = 189$

$\Rightarrow \left(\frac{1 - r \cdot 32}{1-r}\right) = 63 \Rightarrow 1 - 32r = 63 - 63r$

$\Rightarrow 31r = 62 \Rightarrow r = 2$

Now, $2^{n-1} = 32 \Rightarrow n = 6$

Q.57 Answer:

e

Solution:

Let, $L = \lim_{x\to\infty}\left[\frac{e^2}{\left(1+\frac{2}{z}\right)^x}\right]^{\frac{x}{2}}$

Put $\frac{1}{x} = y$

$L = \lim_{y\to 0^+}\left[\frac{e^2}{(1+2y)^{\frac{1}{y}}}\right]^{\frac{1}{2y}}$

$L = e^{\lim_{y\to 0^+}\frac{1}{2y}\ln\left(\frac{e^2}{(1+2y)^{\frac{1}{y}}}\right)} = e^M$ **(let)**

Now, we have, $M = \lim_{y\to 0^+}\frac{1}{2y}\ln\left[\frac{e^2}{(1+2y)^{\frac{1}{y}}}\right]$

$= \lim_{y\to 0^+}\frac{\ln e^2 - \frac{1}{y}\ln(1+2y)}{2y}$

$= \lim_{y\to 0^+}\frac{2y - \ln(1+2y)}{2y^2}$

$= \lim_{y\to 0^+} \frac{2y-\left(2y-\frac{2y^2}{2}+\frac{8y^3}{3}-\frac{16y^4}{4}+\cdots\right)}{2y^2}$

$= \lim_{y\to 0^+} \frac{2}{2}-\frac{4y}{3}+\frac{8y^2}{4}-\ldots\ldots = 1$

Hence, $L = e^1 = e$

Q.58 Answer:

d

Solution:

Since $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ **are the roots of the equation Let us consider the sum of the roots**

$\alpha_1, \alpha_1, +\ldots\ldots\ldots+\alpha_6 = 12\ldots.. (1)$

The product of roots would be

$\alpha_1, \alpha_2, \ldots\ldots\alpha_6 = 64\ldots..(2)$

If we apply arithmetic and geometric mean to equation roots

A.M > G.M

$\alpha_1+\alpha_2+\ldots\ldots+\alpha_6 \geq \quad (\alpha_1, \alpha_2\ldots\ldots+\alpha_6)^{1/6}$

$\frac{12}{6} \geq (\alpha_1, \alpha_2, \ldots\ldots\ldots)^{1/6}$

$2^6 \geq \frac{12}{6} \geq (\alpha_1, \alpha_2, \ldots\ldots\alpha_6)$

$\alpha_1, \alpha_2, \ldots\ldots\alpha_6) \leq 64$

$(\alpha_1, \alpha_2, \ldots\ldots\alpha_6) = 64$

The roots product to equality if all the roots are equal

$\Rightarrow (\alpha_1, \alpha_2, \ldots\ldots\alpha_6) = 2$

Hence the polynomial

$x^6-12x^5+bx^4+cx^3+dx^2+ex+64 = 0$

Since all the root are 2

Hence the maximum of the d **is greatest absolute value to cancel out the polynomial to zero.**

Q.59 Answer:

$\frac{5}{39}$

Solution:

Let, the event $A = 1^{\text{st}}$ **ticket drawn is a perfect square and**

the event $B = 2^{\text{nd}}$ **ticket drawn is a perfect square.**

Thus, $P(A) = \frac{6}{40}$ **and** $P(A\cap B) = \frac{{}^6C_2}{{}^{40}C_2} = \frac{6\times 5}{40\times 39}$

$\Rightarrow P\left(\frac{B}{A}\right) = \frac{P(B\cap A)}{P(A)}$

$= \frac{6\times 5}{40\times 39}\times\frac{40}{6} = \frac{5}{39}.$

Q.60 Answer:

2

Solution:

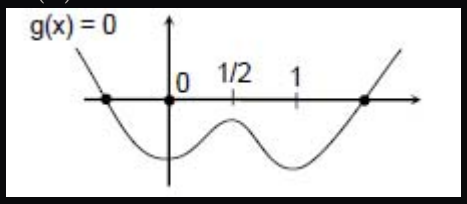
Number of point of

$\frac{dy}{dx} = 60x^5 - 120x^4 + 60x^3 - 80x$

$= 20x\left(3x^4 - 6x^3 + 3x^2 - 4\right) = 0$

$g(x) = 3x^4 - 6x^3 + 3x^2 - 4$

$g'(x) = 12x^3 - 18x^2 + 6x = 0$



$g(x)$ **have two zeros on less than zero and other greater than zero Hence given function will have two point of minima and one point of maxima.**

Q.61 Answer:

e

Solution:

We have, $\mathrm{I}_1 = \int_0^1 e^{x^2} dx$ **and** $\mathrm{I}_2 = \int_0^1 2x^2 e^{x^2} dx$

Now, $\mathrm{I}_2 = \int_0^1 2x^2 e^{x^2} dx$

$= \int_0^1 \left(2xe^{x^2}\right) x dx$

Using integration by parts,

$\mathrm{I}_2 = \left[xe^{x^2}\right]_0^1 - \int_0^1 e^{x^2} dx$

$\mathrm{I}_2 + \mathrm{I}_1 = e$

Q.62 Answer:

$\tan\theta = \frac{-1}{4}$

Solution:

$\mathrm{Arg}(z - 5i) = \frac{-\pi}{4}$ **is a ray emanating from** $(0, 5)$ **and making an angle** $\frac{\pi}{4}$ **from the** x**-axis in clockwise direction.**

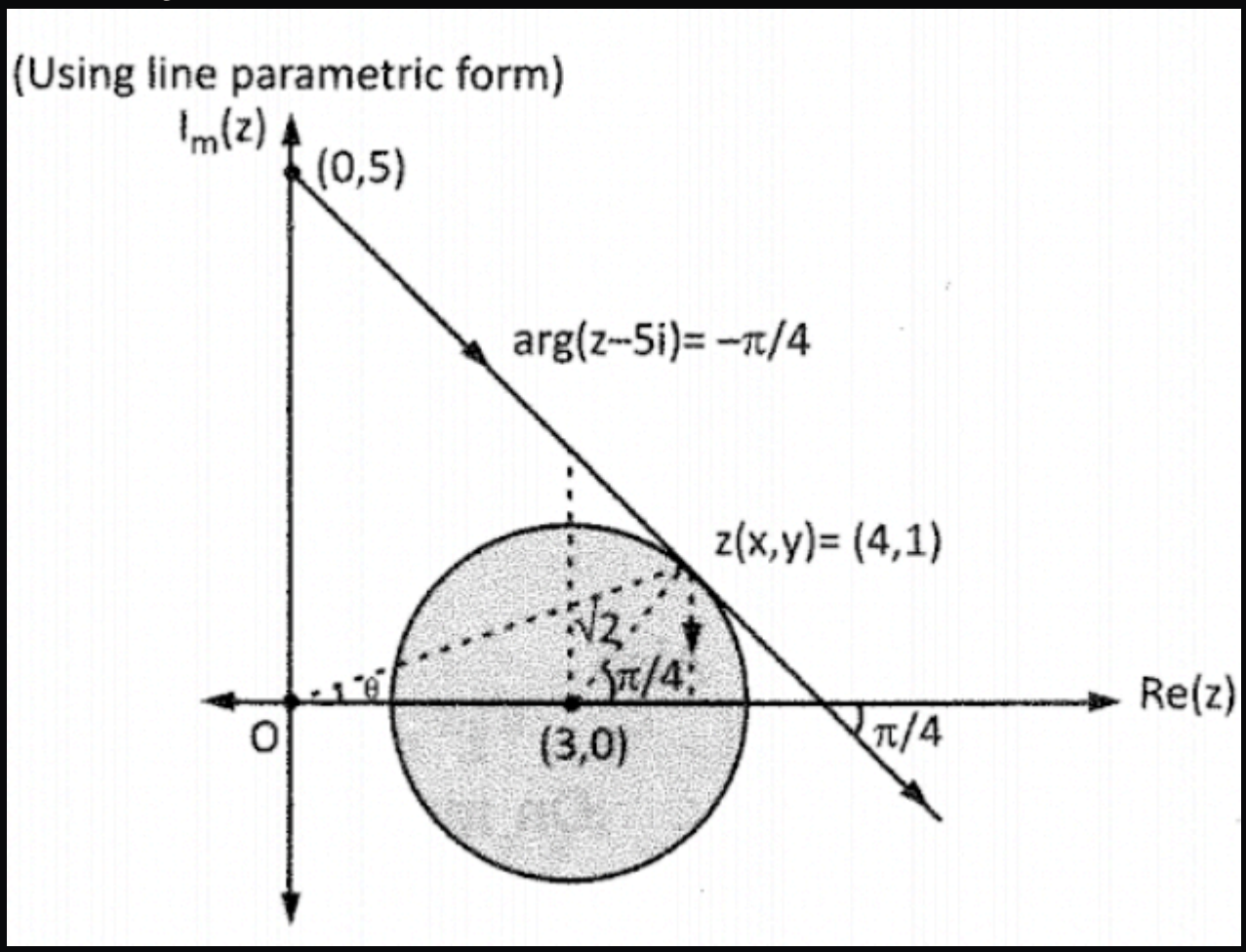
$\Rightarrow \quad \arg[x + i(y - 5)] = \frac{-\pi}{4} \Rightarrow \tan^{-1}\left(\frac{y-5}{x}\right) = \frac{-\pi}{4}$

$\Rightarrow \quad y = 5 - x \Rightarrow \quad x + y = 5$

Also, $\quad |z - 3| \leq \sqrt{2} \Rightarrow |(x - 3) + iy| \leq \sqrt{2}$

$\Rightarrow \quad (x - 3)^2 + y^2 \leq 2$

$\Rightarrow \quad x + y = 5$ **touches the circle.**



Let z_0 **be the complex number corresponding to point of contact.**

Let $z_0 = x_0 + iy_0$

$x_0 = 3 + \sqrt{2}\cos\frac{\pi}{4} = 4$

$y_0 = \sqrt{2}\sin\frac{\pi}{4} = 1$

Now, $z_0 = 4 + i$

$\tan\theta = \frac{1}{4}$

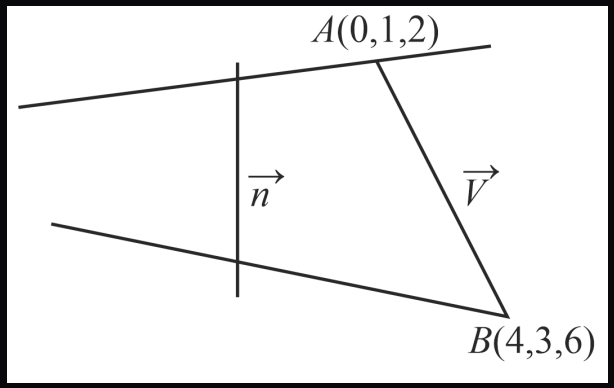
Also, $|z_0| = 17$

and $|z_0 - 5i| = |4 + i - 5i| = 4|1 - i| = 4\sqrt{2}$

Q.63 Answer:

6

Solution:



Vector perpendicular to both $\overrightarrow{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & -1 \\ 1 & 0 & 2 \end{vmatrix}$

$= \hat{i}(-2) - \hat{j}(6+1) + \hat{k}(0+1)$

$= -2\hat{i} - 7\hat{j} + \hat{k}$ says $2\hat{i} + 7\hat{j} - \hat{k}$

now $\overrightarrow{V} = \overrightarrow{AB} = 4\hat{i} + 2\hat{j} + 4\hat{k}$

now $S.D. = \left| \frac{\overrightarrow{V} \cdot \overrightarrow{n}}{|\overrightarrow{n}|} \right| = \left| \frac{8+14-4}{\sqrt{54}} \right| = \frac{18}{\sqrt{54}} = \frac{18}{3\sqrt{6}} = \frac{6}{\sqrt{6}}$

$\therefore SD = \sqrt{6} \Rightarrow d = 6$

Q.64 Answer:

$4e^4 - 24e^2 + 35 = 0$

Solution:

Let equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore$ It passes through $(4, -2\sqrt{3}) \Rightarrow \frac{16}{a^2} - \frac{12}{b^2} = 1$

$\Rightarrow 16 - 12 \times \frac{a^2}{b^2} = a^2$ ...(1)

Equation of directrix is $x = \frac{a}{e} = \frac{4}{\sqrt{5}}$, given in question.

$\Rightarrow a^2 = \frac{16}{5}e^2$ ...(2)

And we know that $b^2 = a^2\left(e^2 - 1\right)$

$\Rightarrow \frac{b^2}{a^2} = e^2 - 1$ ...(3)

$\therefore$ From (1), (2) & (3)

$16 - \frac{12}{e^2-1} = \frac{16}{5}e^2$

$\Rightarrow 16e^2 - 16 - 12 = \frac{16e^2}{5}\left(e^2 - 1\right).$

$\Rightarrow 4e^4 - 24e^2 + 35 = 0$

Q.65 Answer:

${}^nC_r\left(3^{n-r} - 2^{n-r}\right)$

Solution:

We have $(x+3)^{n-1} + (x+3)^{n-2}(x+2) + (x+3)^{n-3}$

$(x+2)^2 + \ldots + (x+2)^{n-1}$

$= \frac{(x+3)^n - (x+2)^n}{(x+3)-(x+2)} = (x+3)^n - (x+2)^n$

$\left[\because \frac{x^n - a^n}{x-a} = x^{n-1} + x^{n-2}a^1 + x^{n-3}a^2 + \ldots + a^{n-1}\right]$

Therefore, coefficient of x^r in the given expression is,

$=$ coefficient of x^r in $\left[(x+3)^n - (x+2)^n\right]$

$= {}^nC_r 3^{n-r} - {}^nC_r 2^{n-r} = {}^nC_r\left(3^{n-r} - 2^{n-r}\right).$

Q.66 Answer:

$a < c < b$

Solution:

$D = B^2 - 4AC$
$= (2(a + b - 2c))^2 - 4(a - b)^2$
$= 4\left\{(a + b - 2c)^2 - (a - b)^2\right\}$
$= 4(a + b - 2c - a + b)(a + b - 2c + a - b)$
$= 4(2b - 2c)(2a - 2c)$
$= 16(b - c)(a - c)$
$= 16(c - b)(c - a)$
If c lies between a and b, then D is negative. Hence, the roots will be imaginary and the graph will be entirely above the x -axis as the coefficient of x^2 is positive.

Q.67 Answer:

$\left(\frac{29}{10},\ 2\right)$

Solution:

$\phi\ (\mathrm{x},\ \mathrm{y}) = \mathrm{x}^2 + \mathrm{y}^2 + 2\mathrm{gx} + 2\mathrm{fy} + \mathrm{c} = 0$
$\phi\ (0,\ \lambda)\ = \lambda^2 + 2\mathrm{f}\lambda + \mathrm{c} = 0 = (\lambda - 2)^2$
$+ 2\mathrm{f} = -4,\ \mathrm{c} = 4$
$\mathrm{f} = -2$ **and** $c = 4$
$\phi(\lambda,\ 0)\ = \lambda^2 + 2\mathrm{g}\lambda + 4 = 0 = \left(\lambda - \frac{4}{5}\right)(\lambda - 5)$
$- 2\mathrm{g} = \frac{4}{5} + 5$
$g = -\left(\frac{29}{10}\right)$,
Thus,
Centre is $(\text{-g},\ -\mathrm{f}) \Rightarrow \left(\frac{29}{10},\ 2\right)$

Q.68 Answer:

two

Solution:

$\because (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x$
Now, from graphs of $\sin^{-1} x, \operatorname{cosec} x$



Clearly, both graph intersects at two points
$\therefore$ two solutions

Q.69 Answer:

5

Solution:

$\because k \in$ odd integer

$f(k) = k + 1$ which is even

$\therefore f(f(k)) = \frac{k+1}{2}$

If $\frac{k+1}{2}$ is odd $\Rightarrow 33 = \frac{k+1}{2} + 1 \Rightarrow k = 63$ not possible (because $\frac{k+1}{2}$ is even)

If $\frac{k+1}{2}$ is even $\Rightarrow 33 = \frac{\frac{k+1}{2}}{2} \Rightarrow 33 = \frac{k+1}{4} \Rightarrow k = 131$

Sum of digits $= 5$

Q.70 Answer:

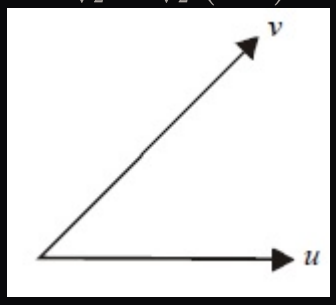
3

Solution:

$\overrightarrow{u} . n = 0 = \overrightarrow{v} . \overrightarrow{n}$

now $\overrightarrow{n} = \pm \frac{\overrightarrow{u} \times \overrightarrow{v}}{|\overrightarrow{u} \times \overrightarrow{v}|}$

$= \pm \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \left(-2\widehat{k}\right) = \pm \widehat{k}$



Now $\overrightarrow{w} . \overrightarrow{n} = |(i + 2j + 3k).(-k)|$

$= |-3| = 3$

Q.71 Answer:

—

Solution:

$[x][y] = 3$

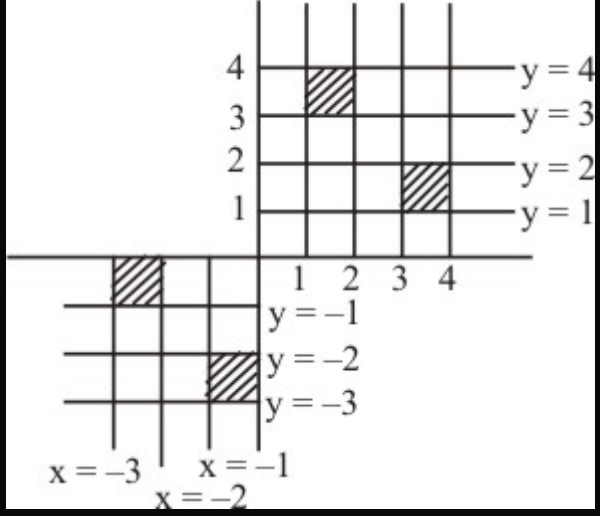
Here four cases arise

$(i)[x] = 1 \ \& \ [y] = 3 \ \Rightarrow 1 \leq x < 2 \ \& \ 3 \leq y < 4$

$(ii)[x] = 3 \ \& \ [y] = 1 \ \Rightarrow 3 \leq x < 4 \ \& \ 1 \leq y < 2$

$(iii)[x] = -3 \ \& \ [y] = -1 \ \Rightarrow -3 \leq x < -2 \ \& \ -1 \leq y < 0$

$(iv)[x] = -1 \ \& \ [y] = -3 \ \Rightarrow -1 \leq x < 0 \ \& \ -3 \leq y < -2$



Required area = Area of shaded region = 4

Q.72 Answer:

—

Solution:

$O(0,0), A(1,0) \& B(-1,0)$

Let $P(x,y)$

$\overrightarrow{PA} = (1-x)\hat{i} - \hat{y}$

$\overrightarrow{PB} = -(1+x)\hat{i} - \hat{y}$

$\overrightarrow{PA} \cdot \overrightarrow{PB} + 3\overrightarrow{OA} \cdot \overrightarrow{OB} = 0$

$\Rightarrow (x^2 - 1) + y^2 - 3 = 0$

$x^2 + y^2 = 4$

$|\overrightarrow{PA}||\overrightarrow{PB}|$

$= \sqrt{(x-1)^2 + y^2}\sqrt{(x+1)^2 + y^2}$

$= \sqrt{5-2x} \cdot \sqrt{5+2x}$

$= \sqrt{25 - 4x^2}, \quad x \in (-2,2)(\text{ from } (1))$

$\text{so } M = 5,\ m = 3 \Rightarrow M^2 + m^2 = 25 + 9 = 34$

Q.73 Answer:

—

Solution:

Since orthocentre and circumcentre both lies on y-axis

$\Rightarrow$ **Centroid also lies on y-axis**

$\Rightarrow \Sigma \cos\alpha = 0$

$\cos\alpha + \cos\beta + \cos\gamma = 0$

$\Rightarrow \cos^3\alpha + \cos^3\beta + \cos^3\gamma = 3\cos\alpha\cos\beta\cos\gamma$

$\therefore \frac{\cos 3\alpha + \cos 3\beta + \cos 3\gamma}{\cos\alpha\cos\beta\cos\gamma}$

$= \frac{4\left(\cos^3\alpha+\cos^3\beta+\cos^3\gamma\right)-3\left(\cos\alpha+\cos\beta+\cos\gamma\right)}{\cos\alpha\cos\beta\cos\gamma}$

$= 12$

Q.74 Answer:

—

Solution:

$|A-nI| = \begin{vmatrix} 1-n & a^3 & 0 \\ 0 & 1-n & b \\ c^3 & 0 & 1-n \end{vmatrix} = 0$

$\Rightarrow (1-n)\left[(1-n)^2\right] - a^3\left[-b^3c^3\right] = 0$

$\Rightarrow -(n-1)^3 + a^3b^3c^3 = 0$

$\Rightarrow (n-1)^3 = (abc)^3$

$\Rightarrow n = 1+abc, 1+abcw, 1+abcw^2$

Sum of non real roots $= 2+abc\left(w+w^2\right)$

$= 2-abc$

$k = 2, m = 1$

$\Rightarrow k+m = 3$

Q.75 Answer:

—

Solution:

In the string **$abacababababcdced$// there are five** $a's$**, four** $b's$**, three** $c's$**, two** $d's$ **and one** e

Case(I): 5 **Unique letters** $= {}^5C_5 = 1$

Case(ii): 2 **Same and** 3 **Different** $= {}^4C_1 \times {}^4C_3 = 16$

Case(iii): 2 **Same,** 2 **More Same and** 1 **Unique**$= {}^4C_2 \times {}^3C_1 = 18$

Case(iv): 3 **same,** 2 **Unique** $= {}^4C_2 \times {}^3C_1 = 18$

case (v): 3 **same,** 2 **same** $= {}^3C_1 \times {}^3C_1 = 9$

Case(vi): 4 **same,** 1 **unique** $= {}^2C_1 \times {}^4C_1 = 8$

case(vii): all 5 **same** $= 1$

Hence, total number of ways $= 71$